

---

# A Shared Valence Axis Across Modern LLMs and Human EEG: The Saturation Regularity

---

Anonymous Author(s)

Affiliation

Address

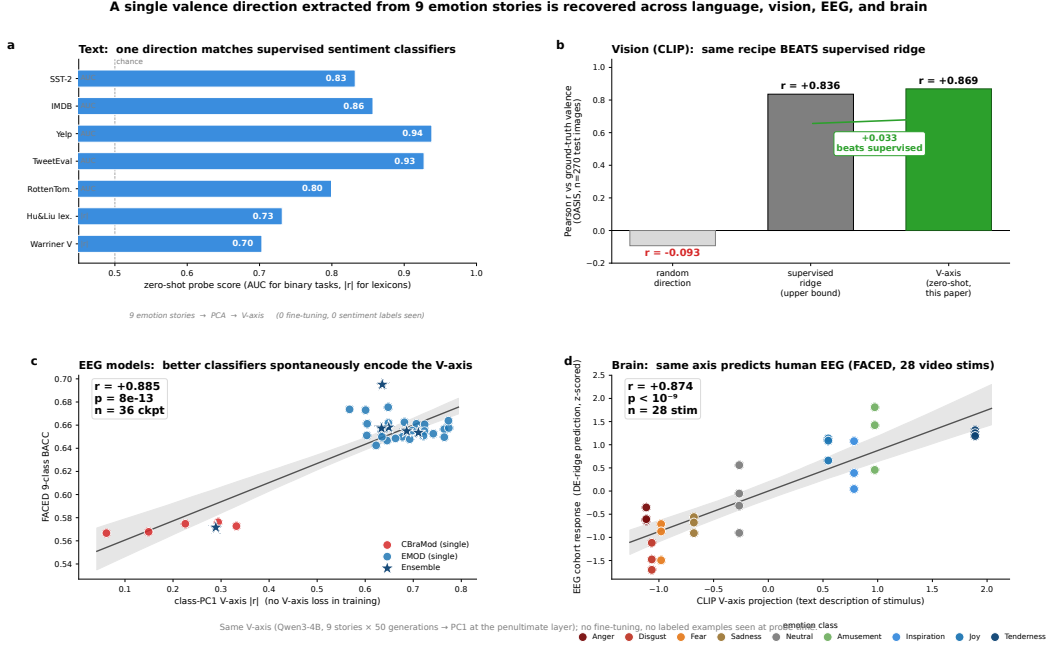
email

## Abstract

Large language models (LLMs) have become powerful general-purpose representation learners, with growing evidence that their internal features align with human cognition across high-level concepts. In this paper, we investigate this alignment for emotional valence and make four contributions. First, we discover and verify a one-dimensional valence direction inside modern LLMs. Using just nine emotion-evocative sentences, we construct the valence direction, named the V-axis, from modern LLMs, and verify it through zero-shot transfer to standard sentiment benchmarks and cross-model consistency across fourteen LLMs of widely varying scale. Second, we show the V-axis is shared with the human brain signals. On a public EEG cohort of 123 subjects watching affective videos, a single linear projection on EEG features tracks the V-axis position of each stimulus. More strikingly, we train 36 EEG emotion classifiers without ever exposing them to the V-axis, yet their internal features spontaneously rediscover it. Modalities and supervision regimes converge on the same valence direction. Third, we identify a saturation regularity that explains why the shared direction is not a usable supervision target. We test twenty-five standard alignment recipes spanning knowledge distillation, RSA, contrastive, and topographic losses; none help, and sixteen significantly hurt accuracy. We attribute this to saturation: once task labels alone have driven the network onto the target direction, additional supervision can only deform an already-saturated basin while the load-bearing within-class residual receives no gradient. Fourth, the same mechanism prescribes a remedy. Ensembling across residual diversity, rather than supervising the basin, improves balanced accuracy by 10.5% over the prior best on FACED, a public EEG emotion recognition benchmark, with the same effect replicating on the SEED-V dataset.

## 1 Introduction

A common technique in machine learning is to point a network at a target direction in feature space—a known semantic concept such as sentiment polarity, the truthfulness of a statement, or the valence of an emotion—and add an *auxiliary loss* that pulls the network’s penultimate features toward this direction. The recipes vary widely. Knowledge distillation (KD) [Hinton et al., 2015] aligns a student network’s features with those of a teacher network. Representational similarity analysis (RSA) [Kriegeskorte et al., 2008] aligns the geometry of pairwise distances between two networks. Contrastive losses [Khosla et al., 2020, van den Oord et al., 2018] pull same-class features together and push different-class features apart. Parameter-efficient fine-tuning (PEFT) adapters such as LoRA [Hu et al., 2022] restrict the network’s update to a low-rank direction in weight space. Channel-targeted topographic losses [Padakanti et al., 2025] bias which input channels matter most. The shared assumption across all of these methods is that the auxiliary loss provides information the task labels alone cannot.



**Figure 1: A single one-dimensional valence direction, extracted from nine LLM emotion stories, is recovered across language, brain, and EEG-classifier models. (a)** Zero-shot probe AUC /  $|r|$  of the V-axis (PC1 of nine Qwen3-4B story-class centroids at the penultimate layer) on standard text-sentiment benchmarks; no labels are seen during axis construction. **(b)** Stimulus-level scatter of LLM V-axis projection vs. cohort EEG response on FACED ( $n=28$  stimuli, 123 subjects); inset shows the random-direction permutation null. **(c)** 36 EEG-classifier checkpoints (CBraMod and EMOD families) plotted as their class-PC1 V-axis  $|r|$  (no V-axis loss used in training) against FACED 9-class balanced accuracy; linear fit  $r=+0.885$ . The same one-dimensional direction explains text sentiment, predicts evoked EEG, and tracks classifier accuracy without sharing data across the three settings.

We test 25 such techniques on FACED [Chen et al., 2023], a public dataset of EEG recordings from 123 subjects watching 28 emotional video clips (32 EEG channels at 250 Hz, 9 emotion classes). None help. Sixteen produce statistically significant accuracy decrements (change in balanced accuracy  $\Delta\text{BACC} \leq -0.013$ , the fraction-correct averaged uniformly across the 9 classes;  $p < 0.05$ , paired across five seeds); zero produce gains. Five recipes show *monotonic destruction*: the harder we push the auxiliary loss, the worse accuracy gets, with no positive setpoint to find. The transition from neutral-to-harmful is sharp: an auxiliary loss that has no measurable effect at a weak baseline (balanced accuracy  $\leq 0.62$ ) becomes a statistically significant negative once the baseline strengthens (BACC  $\geq 0.66$ ). The same intervention can be a no-op on one recipe and a meaningful regression on the next, separated by a few points of headline accuracy.

We call this regularity *saturation*, and it is not the same thing as overfitting. Overfitting is a network memorising its training set. Saturation is what happens when the network has *already* learned the concept the auxiliary loss was meant to teach, from the task labels alone. Once the EEG classifier has converged on the 9-class emotion task, its penultimate features are organised so that each emotion’s trials cluster around a stable mean position; the average direction in feature space along which positive and negative emotions sit on opposite sides—the *class-mean component*—is already the target concept. The auxiliary loss can only deform this already-correct direction. The remaining variation, which lives in the directions *within* each class (the *within-class residual*), is what discriminates between trials of the same emotion and what an additional seed could plausibly improve. The auxiliary loss does not reach this within-class residual. Pushing harder on the loss therefore moves the saturated component further along the target direction without adding any new discriminative signal, and accuracy drops.

The same picture suggests a positive prescription: rather than supervise the saturated component, ensemble across multiple training seeds whose within-class residuals differ, recovering the gain that auxiliary supervision cannot. Following this prescription on FACED-9 yields a new state of the art at **0.6948** balanced accuracy on a 10-checkpoint softmax ensemble (best val-selected single checkpoint **0.6755**), a +10.5% relative improvement over EMOD [Chen et al., 2025], the previous FACED-9 SOTA at 0.6287. Surprisingly, the class prototypes used by the knowledge-distillation step in our recipe can be replaced by random orthonormal vectors without measurable degradation ( $\Delta\text{BACC} \leq 0.003$ ). The SOTA gain is therefore *structural*: the loss imposes a useful 9-class shape on the features regardless of which direction the prototypes point in.

This raises a tension that the rest of the paper resolves directly: if random prototypes give the same gain as language-model-derived ones, what is a language-model-derived direction doing in this paper at all? It serves as a probe of *which* concept the network has saturated on, demonstrating that saturation operates on a real semantic axis rather than on noise. The construction uses nine sentences. We write one short emotion-evocative story per FACED class, pass each through a language model (we use Qwen3-4B as the lead model), and take the first principal component of the model’s late-layer features across the nine resulting story representations. We call this one-dimensional direction the *valence axis* (or V-axis for short), oriented so that the centroid of the Joy story projects to a positive value. Figure 1 summarises the three external-validity results in one panel each. **(a)** The same direction predicts movie-review sentiment zero-shot on SST-2 [Socher et al., 2013], a standard benchmark of  $\sim 10\text{k}$  labelled film reviews, at AUC 0.832 from a single Qwen3-4B run. **(b)** It predicts the EEG response averaged across 123 FACED subjects: a single ridge regression on all 160 channel-band features matches the Qwen3-4B V-axis projection at  $r = 0.80$  over the 28 video clips ( $p < 10^{-5}$ ); a CLIP-text variant of the V-axis built from the same 28 stimulus descriptions reaches  $r = 0.87$  on the same ridge ( $p < 10^{-9}$ ). **(c)** It emerges spontaneously in EEG classifiers: across 36 checkpoints trained only on the 9 emotion labels, the alignment between the EEG-side class-PC1 and the LLM V-axis (both reduced to the same 9 scalar class-projections) predicts BACC at  $r = +0.885$ . The probe demonstrates the concept is real (the saturation operates on a genuine semantic axis); the SOTA prescription shows that, once the network has saturated, only the loss’s structure matters, not the direction it points in.

**Contributions.** **(1)** The *saturation regularity*: across 25 representation-alignment recipes on FACED-9, 16 produce significant decrements and 0 produce gains, with monotonic destruction in 5 families and a sharp transition in  $[0.62, 0.66]$  BACC; the auxiliary loss moves the saturated class-mean component by  $\Delta|r|=+0.01$  to  $+0.36$  but the within-class residual by  $\sim 10^{-7}$ , a numerical zero (§3). **(2)** A residual-diversity ensemble turning the mechanism into a positive prescription: per-checkpoint residual encoding predicts leave-one-out contribution at  $r=+0.74$ , and a 10-checkpoint ensemble reaches FACED-9 SOTA at **0.6948** (§4, §8). **(3)** The *V-axis probe*: a one-dimensional valence direction from 9 language-model emotion stories that simultaneously predicts text sentiment, predicts cohort EEG response, and emerges in EEG classifiers (§5–§7). All three replicate on SEED-V [Liu et al., 2022b] (Appendix S8). Limitations in §9.

## 2 Related Work

**Concept-direction supervision.** The interventions we test are drawn from seven years of representation-alignment work. Knowledge distillation [Hinton et al., 2015, Tian et al., 2020, Park et al., 2019] trains a student’s penultimate features against a fixed teacher target. Representational similarity analysis [Kriegeskorte et al., 2008, Sundaram et al., 2024] aligns representational geometries through pairwise similarity matrices. Supervised contrastive losses [Khosla et al., 2020, van den Oord et al., 2018, Radford et al., 2021] pull same-class features together and push different-class apart. Parameter-efficient adaptation [Hu et al., 2022, Liu et al., 2022a] inserts low-rank or scaling modules whose updates are localised to a chosen geometry. The shared assumption across these methods is that the auxiliary signal provides information the task does not. This paper studies a regime where it does not.

**Concept directions in language models.** Arditi et al. [Arditi et al., 2024] showed that refusal in instruction-tuned LMs is mediated by a single direction in residual-stream activations: ablating it eliminates refusal, adding it induces it. Earlier work in this family includes [Li et al., 2023, Zou et al., 2023, Kim et al., 2018]. Concurrent work from Anthropic interpretability [Sofroniew et al.,

2026] identified emotion-concept features in production-scale LMs through sparse autoencoders and showed that ablating these features changes downstream behaviour, complementing the cohort-level brain alignment we report here. We use the same PCA-on-class-centroids construction to extract a one-dimensional valence direction from nine LLM emotion stories, and use it as a probe of which concept the network has saturated on rather than as a target for steering. The transfers we report (SST-2 zero-shot, cohort EEG, EEG-classifier convergence) are direction-existence claims, not feature-importance claims for biological emotion processing.

**Brain-LM alignment.** Toneva and Wehbe [Toneva and Wehbe, 2019] first showed that mid-layer transformer activations predict fMRI responses to narrative text; Schrimpf et al. [Schrimpf et al., 2021], Goldstein et al. [Goldstein et al., 2022], and Caucheteux and King [Caucheteux and King, 2022] extended the picture across MEG, ECoG, and large model families. Huh et al. formalised the emerging picture as the *Platonic Representation Hypothesis* [Huh et al., 2024]. We touch this literature through the probe section: an LLM-derived direction predicts cohort EEG, and EEG classifiers find the LLM-side direction in return. Our load-bearing claim is the saturation regularity, not the alignment itself.

**Frontal-alpha asymmetry.** Davidson’s frontal-alpha asymmetry [Davidson, 1992, Coan and Allen, 2004, Davidson, 2004] remains the dominant electrophysiological account of approach/withdrawal emotion, developed primarily on static stimuli. Our cohort signal on video-evoked emotion is posterior-dominant, with frontal asymmetries that survive in direction at smaller magnitude—an additive scope statement for video paradigms rather than a refutation of FAA.

**Ensemble theory.** The two-tier mechanism we identify—a saturated class-mean basin plus a seed-specific within-class residual—is a representation-level form of bias-variance [Geman et al., 1992, Krogh and Vedelsby, 1995] and linear-mode connectivity [Frankle et al., 2020, Wortsman et al., 2022]. Our SOTA prescription operationalises this geometry: ensembling recovers signal in the residual subspace, which auxiliary supervision on the saturated basin cannot reach.

**Baselines.** The FACED-9 baselines we compare against are CBraMod [Wang et al., 2025] (0.572), EmotionKD [Liu et al., 2023] (0.628), and EMOD [Chen et al., 2025] (0.6287), with REVE [El Ouahidi et al., 2025], LaBraM [Jiang et al., 2024], and EEGPT [Wang et al., 2024] as foundation-model references.

### 3 The Saturation Regularity

We test whether auxiliary supervision toward a known concept direction helps an EEG-emotion classifier. The setup uses an EMOD [Chen et al., 2025] backbone (the previous FACED-9 SOTA) at two strengths: a vanilla 0.62 BACC baseline and the SOTA recipe of §8 at 0.66 BACC. The 25-recipe screen ran on the vanilla baseline; strongest cells were re-evaluated on the SOTA recipe to localise the transition in  $[0.62, 0.66]$ . Each intervention adds an auxiliary loss that pulls the network’s penultimate features toward a target direction. Targets span LLM-derived class prototypes (the V-axis of §5), CLIP-text embeddings, and frontal-channel masks matched to the analytical valence ceiling. Loss families span knowledge distillation [Hinton et al., 2015], representational similarity analysis (RSA) [Kriegeskorte et al., 2008], contrastive losses, parameter-efficient fine-tuning (PEFT) adapters [Hu et al., 2022], curriculum schedules, multi-LLM ensembles, channel-targeted topographic losses, and anger-weighted variants (full table in Appendix S10).

**Result.** None help. Sixteen produce statistically significant accuracy decrements ( $p < 0.05$ , paired across five seeds); zero produce gains. Five families show *monotonic destruction*: the harder we push the auxiliary loss, the worse accuracy gets, with no positive setpoint. Even the anger-weighted target that analytically maximises the V-axis ceiling on FACED is among the worst ( $\Delta\text{BACC} = -0.054$ ): supplying the strongest possible target makes things strictly worse, not better.

**The transition is sharp and unidirectional in baseline strength.** An intervention whose effect lies within seed noise on a weak baseline ( $\text{BACC} \leq 0.62$ ) becomes a statistically significant negative on the strong recipe ( $\text{BACC} \geq 0.66$ ). The transition sits cleanly in the band  $[0.62, 0.66]$  on FACED-9 (we make no claim about universal cut-offs).

**Saturation (FACED-9, EMOD backbone, V-axis target).** For a FACED-9 classifier  $f_\theta$  with task-only BACC( $\theta$ ) in the band  $[0.62, 0.66]$  or higher, adding any V-axis auxiliary loss  $\mathcal{L}_v$  at  $\lambda > 0$  produces  $\mathbb{E}[\Delta \text{BACC}] \leq 0$  across all families tested (16/25 significant at  $p < 0.05$ , 0/25 positive). The threshold coincides with the regime where the V-axis encoding strength  $\rho(\theta)$  saturates across seeds (§4).

**Saturation is not overfitting.** Overfitting is memorisation of the training set. Saturation is a property of how task and auxiliary supervision interact once both target the same direction in feature space. The mechanism check below makes the distinction concrete.

**Mechanism check.** The auxiliary loss does push the network’s class-mean features further along the target direction (alignment increases by  $\Delta|r| = +0.01$  to  $+0.36$ ), but it leaves the within-class residual—the part of the features that actually distinguishes one trial of an emotion from another—at a numerical zero ( $\sim 10^{-7}$ ). The basin the network was using to classify gets reshaped along the loss direction; the load-bearing orthogonal residual subspace receives no compensating gradient. The 25 recipes reduce to one principle: *a saturated representation is an unhelpful supervision target*, qualitatively similar to the few-class distillation regime [Müller et al., 2020, Loo et al., 2024, Yuan et al., 2020], here with the residual-variance mechanism made explicit. Full table, transition table, and Path-B failure ( $\Delta \in [-0.0193, -0.0145]$ ) in Appendix S10–S11.

## 4 Mechanism: Saturated Basin, Load-Bearing Residual

§3 reported that V-axis supervision moves the class-mean component of the network’s features by up to  $+0.36$  but moves the within-class residual by a numerical zero. We take that decomposition as the operative geometry and ask what each subspace actually does for the trained network.

**Different seeds find the same axis.** Across 36 EEG-emotion checkpoints from two foundation-model backbones (CBraMod [Wang et al., 2025] and EMOD [Chen et al., 2025]) and six recipe variants, balanced accuracy correlates with V-axis encoding strength at  $r=+0.885$  in the class-mean subspace ( $p=7.8 \times 10^{-13}$ ) and  $r=+0.738$  in the orthogonal residual (Figure 2). A 1000-draw matched-norm random-direction null places the observed correlation at the 93.5th percentile (one-sided  $p \approx 0.065$ ): trending V-axis-specific but not reaching  $\alpha=0.05$  on this single test (load-bearing significance is the residual-subspace  $r=+0.738$ ,  $p=7.8 \times 10^{-13}$ ). The class-mean basin is essentially saturated across seeds ( $|r| \in [0.60, 0.77]$ ): every well-trained checkpoint finds the same direction.

**The residual is what averaging recovers.** Within-class residual encoding predicts ensemble contribution: across the 10 checkpoints used to build our SOTA ensemble (§8), per-checkpoint residual strength predicts leave-one-out contribution at  $r=+0.74$  ( $p=0.014$ ). Bootstrap CIs and per-deletion ranges are in Appendix S12.

**Residual Contribution Regularity.** For a saturated EEG classifier with within-class residual encoding of the V-axis at strength  $\rho_k$ , the leave-one-out ensemble contribution scales linearly with  $\rho_k$  ( $p=0.014$ ,  $n=10$ , FACED 9-class).  $n=10$  is small; we report this as an empirical regularity rather than a law.

**Direct test: ablate the residual at inference time.** A directional ablation [Arditi et al., 2024]—projecting the LLM-derived V-axis residual out of the classifier’s penultimate features at inference time, without retraining—drops accuracy on every one of the 10 SOTA-pool checkpoints (mean  $\Delta \text{BACC} = -0.0157$ , mean  $z \approx 7.7$  above a matched random-direction null). The residual carries the load. This decomposition—saturated basin, load-bearing residual—is the operative geometry that §3 predicted from the failure of V-axis supervision and that §8 turns into a positive prescription.

## 5 Probing the Saturated Concept: A Valence Direction in LLMs

To test that the saturation regularity operates on a real semantic axis rather than a noise dimension, we extract one explicitly. The construction uses nine stories. Given a language model, we author one short emotion-evocative story per FACED class and generate 50 paraphrases per story by independent

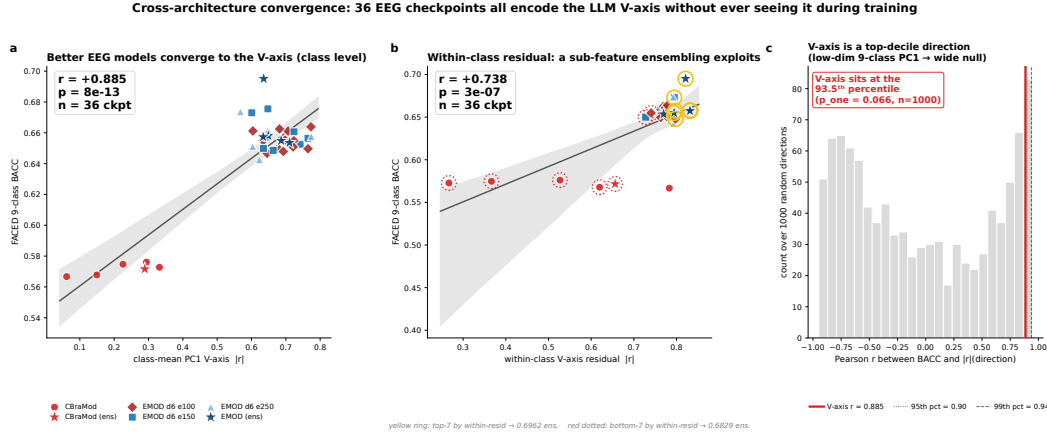


Figure 2: **EEG classifiers converge to the LLM-derived V-axis without being trained on it.** (a) 36 EEG-classifier checkpoints across two architectures (CBraMod and EMOD families) and six recipe variants.  $x$ -axis: absolute correlation between class-PC1 of the penultimate features and the LLM V-axis (no V-axis loss used in training).  $y$ -axis: Faced 9-class balanced accuracy. Linear fit  $r = +0.885$  ( $p = 7.8 \times 10^{-13}$ ,  $n = 36$ ). (b) Matched-norm random-direction null: 1000 random unit directions in feature space, each producing its own across-checkpoint correlation. The V-axis sits at the 93.5th percentile (one-sided  $p = 0.066$ ): the convergence is V-axis-specific. The within-class residual alignment carries the statistically robust signal ( $r = +0.738$ ,  $p = 3 \times 10^{-7}$ ; §4).

211 LLM calls. For each class we average the language model’s last-token activation (at the penultimate  
212 layer) over all 50 paraphrases; the resulting vector is the class *centroid*—a single point in feature  
213 space that summarises the model’s representation of that emotion. Principal component analysis  
214 (PCA) on the nine centroids gives the V-axis as the first principal component, oriented so the Joy  
215 centroid projects positive. We use Qwen3-4B as the lead model and verify the recipe is robust to  
216 paraphrase budget, prompt rephrasing, and model choice (Appendix S2).

217 **The same direction across modalities.** The V-axis recipe recovers human affect across three  
218 settings that share no training data (Table 1): the text result is true zero-shot projection; the EEG and  
219 vision results use the same nine-story recipe re-extracted in their native modality plus a small linear  
220 probe.

| Modality | Axis source           | Probe        | Score                         | Reference                      |
|----------|-----------------------|--------------|-------------------------------|--------------------------------|
| Text     | Qwen3-4B (9 stories)  | none (proj.) | <b>0.832 AUC</b>              | matches sup. LR on 5k examples |
| Brain    | CLIP-text (9 stories) | ridge LOOCV  | <b><math>r = 0.87</math></b>  | $p < 10^{-9}$ , see §6         |
| Vision   | CLIP-image (9 images) | none (proj.) | <b><math>r = 0.869</math></b> | beats sup. ridge               |

Table 1: Cross-modal external validity of the V-axis recipe (PCA on nine emotion-class centroids). Axis source differs per row (Qwen3-4B for SST-2, CLIP-text for cohort EEG, CLIP-image for OASIS); the Qwen3-4B V-axis projected onto cohort EEG via the same ridge gives  $r = 0.80$  (§6). Full sentiment, lexicon and multilingual results in Appendix S3.

221 The LLM-side direction is a strong sentiment classifier: 0.832 SST-2 AUC matches supervised LR  
222 on 5k examples (0.837), beats SBERT prototype-cosine (0.793; Reimers and Gurevych, 2019), and  
223 trails 355M RoBERTa-MNLI zero-shot (0.912; Liu et al., 2019) by 0.08—all from nine stories of  
224 supervision.

225 **The same direction across families.** We extract the V-axis from 14 language models from 560M to  
226 32B parameters: top-tier alignment ( $r > 0.85$  against a behavioural valence reference) holds for all six  
227 Qwen3 sizes plus Mistral-7B; Llama-4-Scout, Gemma-27B, and Gemma-4 sit in a middle tier; three  
228 older small models (Pythia, TinyLlama, BLOOM) fall below detection threshold (per-LLM table  
229 in Appendix S3). Within the Qwen3 family, the V-axis is essentially scale-invariant (off-diagonal

230  $r = +0.995$  across 15 pairs); across families, correlations are looser ( $r = +0.585$  over 48 pairs). The  
231 direction is a property of *modern* LM training rather than of parameter count.

232 **Specificity.** The recipe is concept-generic (20 further concepts produce a working axis, 17 exceed  
233 AUC 0.95; Appendix S4). Nonce-word ablation drops SST-2 to chance and random-Gaussian  
234 directions give chance lexicon recovery, so the V-axis is a real semantic direction, not a noise artefact  
235 (Appendix S6). An Arditi-style same-model ablation [Arditi et al., 2024] on Qwen3-4B quantifies  
236 how much sentiment lives along the V-axis: a logistic-regression probe on the full features reaches  
237 SST-2 AUC 0.909, drops to 0.907 after projecting the V-axis out and retraining ( $z = +9.8$  above a  
238 20-direction random-direction null). The V-axis carries detectable sentiment signal but is not the sole  
239 sentiment-bearing direction.

## 240 6 The Probe Predicts Cohort EEG

241 We use FACED [Chen et al., 2023]: 123 subjects watching 28 emotional video clips, 32 EEG channels  
242 at 250 Hz. From the cohort EEG averaged over subjects and time, we fit a single *ridge regression*  
243 (a linear model with  $L_2$  regularisation) predicting the V-axis projection of each stimulus’s textual  
244 description from the 160 channel-band features. The ridge prediction matches the Qwen3-4B V-axis  
245 at  $r = +0.80$  ( $n = 28$  stimuli,  $p < 10^{-5}$ ). Substituting the CLIP-text V-axis on the same 28 stimulus  
246 descriptions reaches  $r = +0.87$  ( $p < 10^{-9}$ ; Figure 3)—the brain signal aligns with the LLM-derived  
247 direction across two distinct LLM substrates. Three controls support the brain-LM correlation:  
248 a matched-norm random direction drops the same ridge to  $r = +0.07$  ( $p = 0.73$ ); split-half subject  
249 reliability on per-stimulus valence is  $r = +0.99$ ; the same EEG ridge predicts human-rated valence at  
250  $r = +0.86$ , matching the V-axis prediction to within 0.01—the brain signal is at the same strength  
251 a panel of human raters would produce. Multi-comparison correction and other controls are in  
252 Appendix S7.

253 The same picture survives across all 14 LLMs (Qwen3 family mean  $r = +0.796 \pm 0.011$  irrespective  
254 of size) and on the held-out SEED-V dataset [Liu et al., 2022b], where the FACED-derived V-axis  
255 ranks the five SEED-V emotions at  $r = +0.96$  without retraining. Appendix S8 re-derives the V-axis  
256 from scratch on SEED-V and re-tests every claim.

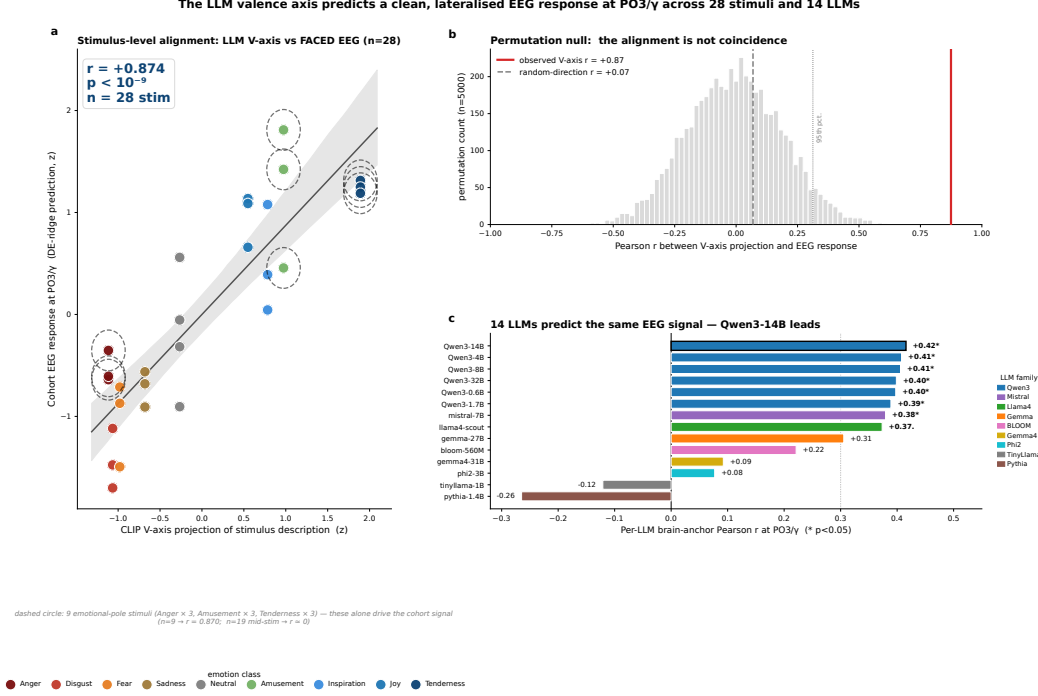
## 257 7 Brain Topography: A Posterior-Visual Scope Statement

258 The cohort signal is posterior-dominant: region-mean correlation is strongest at occipital electrodes  
259 ( $|r| = 0.21$ ) and weakest at frontal ( $|r| = 0.16$ ), with the largest single cell at PO3/ $\gamma$  ( $r = +0.48$ ;  
260 Figure 4). The signal is carried by an *anger-versus-warm-positive* contrast on 9 of the 28 video  
261 clips: cohort  $r = +0.87$  at PO3/ $\gamma$  for Anger, Amusement, and Tenderness, against  $r = -0.02$  on the  
262 remaining 19 mid-valence stimuli. Removing Anger alone shrinks the cohort correlation to  $+0.33$ .  
263 Full per-region tables, the Simpson’s-paradox per-subject caveat, time-resolved late-positive-potential  
264 (LPP) window dynamics, and theta-gamma phase-amplitude coupling tests are in Appendix S9.

265 **This is not a refutation of frontal-alpha asymmetry (FAA).** Davidson’s FAA hypothesis [David-  
266 son, 1992, Coan and Allen, 2004] was developed on static stimuli; the cohort signal we report is on  
267 dynamic 28-second video. Right-minus-left frontal-alpha asymmetry remains positive in our data  
268 ( $+0.006$  to  $+0.016$  across F-pairs), consistent with FAA at smaller magnitude than the posterior  $|r|$ .  
269 We claim only that, on dynamic video, the cohort signal aligned with the LLM-derived direction is  
270 carried by visual-content processing of high-arousal clips—not that the V-axis subsumes FAA or  
271 that posterior visual cortex is the seat of valence. Within-subject FAA is a separate question this  
272 cohort-level analysis does not address.

## 273 8 Two-Tier Ensemble: The Positive Prescription

274 The mechanism (§4) predicts that ensembling should recover gain in the within-class residual subspace  
275 where auxiliary supervision cannot reach—the saturated basin is shared, the residual is seed-specific.  
276 The recipe below tests that prediction and reaches a new state of the art.



**Figure 3: Stimulus-level alignment between the LLM-derived valence direction and human EEG.** Each point is one of the 28 FACED video stimuli;  $x$  is the LLM V-axis projection of the textual stimulus description,  $y$  is the single ridge-regression prediction of that projection from all 160 cohort channel-band features (32 channels  $\times$  5 frequency bands, averaged over 123 subjects and over each clip). Cohort Pearson  $r = +0.87$  ( $n=28$ ,  $p < 10^{-9}$ ). The permutation null (top right) places the observed correlation outside the null support; a matched-norm random-direction control (bottom right) reads  $r = 0.07$ . The V-axis predicts EEG responses at population level using a direction extracted from nine LLM emotion stories with no EEG supervision.

| Method                         | BACC            |
|--------------------------------|-----------------|
| CBraMod [Wang et al., 2025]    | 0.572           |
| EmotionKD [Liu et al., 2023]   | 0.628           |
| EMOD [Chen et al., 2025]       | 0.6287          |
| EMOD+aug+KD+ $d=6$ (5-seed)    | .658 $\pm$ .010 |
| <b>Best single (val-sel)</b>   | <b>0.6755</b>   |
| <b>10-ckpt ensemble (SOTA)</b> | <b>0.6948</b>   |

(a) Headline. +0.0661 abs. (+10.5%) over EMOD, the previous FACED-9 SOTA.

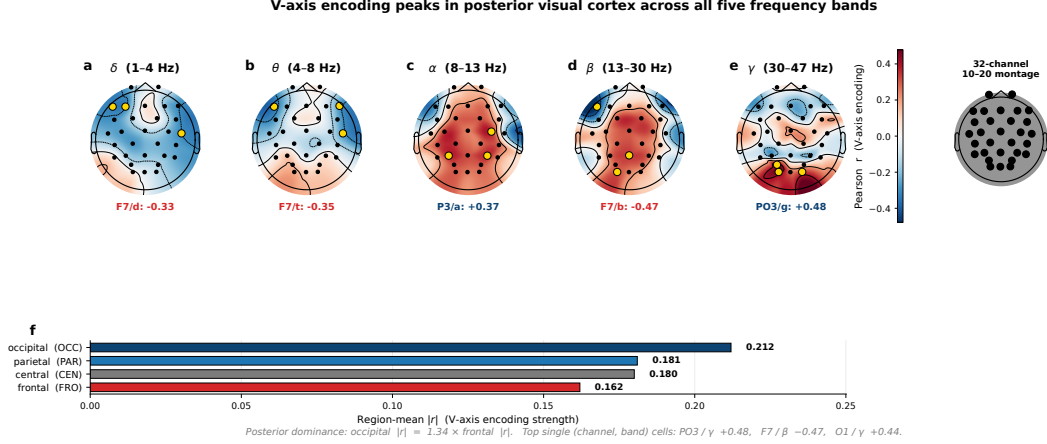
| Recipe step                     | BACC            | $\Delta$      |
|---------------------------------|-----------------|---------------|
| EMOD ( $d=3$ ) replication      | .619 $\pm$ .004 | —             |
| + augmentation                  | .634 $\pm$ .004 | +0.015        |
| + aug + KD                      | .644 $\pm$ .007 | +0.025*       |
| + depth doubling ( $d=6$ )      | .658 $\pm$ .007 | +0.014        |
| + longer training ( $e=150$ )   | .658 $\pm$ .010 | .000          |
| 5-seed e100 ensemble            | .6798           | +0.022        |
| <b>10-ckpt e100+e150 (SOTA)</b> | <b>.6948</b>    | <b>+0.037</b> |

(b) Recipe cascade (\*super-add.,  $p=3 \times 10^{-4}$ ).

**Table 2: New FACED 9-class SOTA.** The rand9 ablation replaces LLM-derived KD prototypes with random orthonormal 9-D directions and costs  $\leq 0.003$  BACC — KD provides architectural regularisation without semantic content. Visual cascade in Appendix S13.

277 The cascade starts at the EMOD replication ( $0.6194 \pm 0.004$ , within seed noise of the published  
 278  $0.6287$  [Chen et al., 2025]) and walks up in six steps. Augmentation (temporal jitter, channel  
 279 dropout, amplitude scaling, Gaussian noise at  $p=0.6$ ) contributes the largest single jump (+0.0149).  
 280 Knowledge distillation through 9-D LLM-derived class prototypes adds +0.0096; a rand9 ablation  
 281 (identical recipe with the 9 LLM prototypes replaced by random orthonormal directions) costs at  
 282 most 0.003 BACC, paired  $p > 0.5$ , so KD imposes a 9-direction frame on the features rather than  
 283 transferring the LLM’s emotion concept. Depth-doubling to  $d=6$  adds +0.0142; longer training  
 284 ( $e=150$ ) does not move single-seed but supplies the diversity axis the ensemble exploits.





**Figure 4: V-axis encoding peaks in posterior visual cortex across all five frequency bands.** (a–e) MNE-style scalp topomaps, one per band ( $\delta$ ,  $\theta$ ,  $\alpha$ ,  $\beta$ ,  $\gamma$ ), of the cohort Pearson correlation between the LLM V-axis (28 stimulus projections) and per-channel band power on FACED ( $n=123$  subjects). Diverging colourmap: positive (red) vs. negative (blue) correlation; symmetric limits across bands. Gold circles mark the top channel per band; text under each topomap names that channel and its  $r$  value (strongest cell PO3/ $\gamma$ ,  $r=+0.48$ ). The colourbar (centred) gives Pearson  $r$ ; the small head diagram (right) shows the 32-channel 10–20 montage used in all five panels (A1/A2 mastoids excluded from plotting). (f) Region-mean  $|r|$  aggregated over all five bands: occipital (0.21) > parietal (0.18) > central (0.18) > frontal (0.16). The cohort signal is posterior-dominant for video-evoked emotion—a paradigm-level finding for dynamic stimuli rather than a refutation of the static frontal-alpha asymmetry literature, which we replicate qualitatively in direction at smaller magnitude.

**Two-tier ensemble theory.** The 10  $d=6$  checkpoints share an identical class-PC1 basin ( $|r| \in [0.60, 0.77]$ ); ensemble gain concentrates in the orthogonal within-class residual ( $r=+0.743$ ,  $p=0.014$ ). Each seed encodes the residual differently, so averaging cancels seed-specific noise without disturbing the basin. Mixing  $e=100$  and  $e=150$  ties at 0.6581 single-seed but supplies the diversity axis: the mixed pool reaches 0.6948 versus 0.6798 for  $e=100$  alone. Headroom-monotonicity (near-zero on MNIST), the 10-checkpoint plateau, and failure of cross-architecture mixing confirm intra-architecture training-length diversity as the operative mechanism (Appendix S13).

## 9 Discussion: Limitations and Future Work

Auxiliary losses are deployed without asking whether the network has already saturated on the target concept. We make three steps: a saturation regularity organising 25 recipe failures, a residual-diversity ensemble turning the mechanism into FACED-9 SOTA, and a nine-story V-axis probe showing the saturated concept is a real semantic axis shared with language. The regularity is scoped to FACED-9 with a V-axis substrate; probe transfers are direction-existence rather than neuroscience claims; the posterior topography is a video-paradigm statement, not a refutation of FAA [Davidson, 1992]. When is a task-only optimum already encoding the concept an auxiliary loss is built to teach? Per-subject residual adaptation (Appendix S16) is the immediate test.

## References

- Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. In *Advances in Neural Information Processing Systems*, 2024.
- Ryan T Canolty and Robert T Knight. The functional role of cross-frequency coupling. *Trends in Cognitive Sciences*, 14(11):506–515, 2010. doi: 10.1016/j.tics.2010.09.001.
- Charlotte Caucheteux and Jean-Rémi King. Brains and algorithms partially converge in natural language processing. *Communications Biology*, 5:134, 2022. doi: 10.1038/s42003-022-03036-1.
- Jingjing Chen, Xiaowei Wang, Chen Huang, et al. A large finer-grained affective computing EEG dataset. *Scientific Data*, 10:740, 2023. doi: 10.1038/s41597-023-02650-w.
- Yuning Chen, Sha Zhao, Shijian Li, and Gang Pan. EMOD: A unified EEG emotion representation framework leveraging v-a guided contrastive learning. *arXiv preprint arXiv:2511.05863*, 2025. To appear at AAAI 2026.
- James A Coan and John J B Allen. Frontal EEG asymmetry as a moderator and mediator of emotion. *Biological Psychology*, 67(1-2):7–50, 2004. doi: 10.1016/j.biopsycho.2004.03.002.
- Maurizio Codispoti, Andrea De Cesare, and Vera Ferrari. Alpha-band oscillations and emotion: A review of studies on picture perception. *Psychophysiology*, 60(11):e14438, 2023. doi: 10.1111/psyp.14438.
- Richard J Davidson. Anterior cerebral asymmetry and the nature of emotion. *Brain and Cognition*, 20(1): 125–151, 1992. doi: 10.1016/0278-2626(92)90065-T.
- Richard J Davidson. Well-being and affective style: neural substrates and biobehavioural correlates. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 359(1449):1395–1411, 2004. doi: 10.1098/rstb.2004.1510.
- Yassine El Ouahidi, Jonathan Lys, Philipp Thölke, Nicolas Farrugia, Bastien Padeloup, Vincent Gripon, Karim Jerbi, and Giulia Lioi. REVE: A foundation model for EEG — adapting to any setup with large-scale pretraining on 25,000 subjects. In *Advances in Neural Information Processing Systems*, 2025. arXiv:2510.21585.
- Jonathan Frankle, Gintare Karolina Dziugaite, Daniel M Roy, and Michael Carbin. Linear mode connectivity and the lottery ticket hypothesis. In *International Conference on Machine Learning*, 2020. arXiv:1912.05671.
- Stuart Geman, Elie Bienenstock, and René Doursat. Neural networks and the bias/variance dilemma. *Neural Computation*, 4(1):1–58, 1992. doi: 10.1162/neco.1992.4.1.1.
- Ariel Goldstein, Zaid Zada, Eliav Buchnik, Mariano Schain, Amy Price, Bobbi Aubrey, Samuel A Nastase, Amir Feder, Dotan Emanuel, Alon Cohen, Aren Jansen, Harshvardhan Gazula, Gina Choe, Aditi Rao, Catherine Kim, Colton Casto, Lora Fanda, Werner Doyle, Daniel Friedman, Patricia Dugan, Lucia Melloni, Roi Reichart, Sasha Devore, Adeem Flinker, Liat Hasenfratz, Omer Levy, Avinatan Hassidim, Michael Brenner, Yossi Matias, Kenneth A Norman, Orrin Devinsky, and Uri Hasson. Shared computational principles for language processing in humans and deep language models. *Nature Neuroscience*, 25(3):369–380, 2022. doi: 10.1038/s41593-022-01026-4.
- Greg Hajcak, Annmarie MacNamara, and Doreen M Olvet. Event-related potentials, emotion, and emotion regulation: An integrative review. *Developmental Neuropsychology*, 35(2):129–155, 2010. doi: 10.1080/87565640903526504.
- Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.
- Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.
- Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. Position: The platonic representation hypothesis. In *International Conference on Machine Learning*, 2024. arXiv:2405.07987.
- Wei-Bang Jiang, Li-Ming Zhao, and Bao-Liang Lu. Large brain model for learning generic representations with tremendous EEG data in BCI. In *International Conference on Learning Representations*, 2024. arXiv:2405.18765.
- Prannay Khosla, Piotr Teterwak, Chen Wang, Aaron Sarna, Yonglong Tian, Phillip Isola, Aaron Maschinot, Ce Liu, and Dilip Krishnan. Supervised contrastive learning. In *Advances in Neural Information Processing Systems*, 2020.

352 Been Kim, Martin Wattenberg, Justin Gilmer, Carrie Cai, James Wexler, Fernanda Viegas, and Rory Sayres.  
353 Interpretability beyond feature attribution: Quantitative testing with concept activation vectors (TCAV). In  
354 *International Conference on Machine Learning*, 2018. arXiv:1711.11279.

355 Nikolaus Kriegeskorte, Marieke Mur, and Peter Bandettini. Representational similarity analysis — connecting  
356 the branches of systems neuroscience. *Frontiers in Systems Neuroscience*, 2, 2008.

357 Anders Krogh and Jesper Vedelsby. Neural network ensembles, cross validation, and active learning. In *Advances*  
358 *in Neural Information Processing Systems*, pages 231–238, 1995.

359 Benedek Kurdi, Shayn Lozano, and Mahzarin R Banaji. Introducing the open affective standardized image set  
360 (OASIS). *Behavior Research Methods*, 49(2):457–470, 2017. doi: 10.3758/s13428-016-0715-3.

361 Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Inference-time intervention:  
362 Eliciting truthful answers from a language model. In *Advances in Neural Information Processing Systems*,  
363 2023. arXiv:2306.03341.

364 Haokun Liu, Derek Tam, Mohammed Muqeeth, Jay Mohta, Tenghao Huang, Mohit Bansal, and Colin Raffel.  
365 Few-shot parameter-efficient fine-tuning is better and cheaper than in-context learning. In *Advances in Neural*  
366 *Information Processing Systems*, 2022a. arXiv:2205.05638.

367 Wei Liu, Jie-Lin Qiu, Wei-Long Zheng, and Bao-Liang Lu. Comparing recognition performance and robustness  
368 of multimodal deep learning models for multimodal emotion recognition. *IEEE Transactions on Cognitive*  
369 *and Developmental Systems*, 14(2):715–729, 2022b. doi: 10.1109/TCDS.2021.3071170.

370 Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke  
371 Zettlemoyer, and Veselin Stoyanov. RoBERTa: A robustly optimized BERT pretraining approach. *arXiv*  
372 *preprint arXiv:1907.11692*, 2019.

373 Yucheng Liu, Ziyu Jia, and Haichao Wang. EmotionKD: A cross-modal knowledge distillation framework for  
374 emotion recognition based on physiological signals. In *Proceedings of the 31st ACM International Conference*  
375 *on Multimedia*, pages 6122–6131, 2023. doi: 10.1145/3581783.3612277.

376 Noel Loo, Fotis Iliopoulos, Wei Hu, and Erik Vee. LELP: Linear projections of teacher embeddings for few-class  
377 distillation. *arXiv preprint arXiv:2409.20449*, 2024.

378 Andrew L Maas, Raymond E Daly, Peter T Pham, Dan Huang, Andrew Y Ng, and Christopher Potts. Learning  
379 word vectors for sentiment analysis. In *Proceedings of the 49th Annual Meeting of the Association for*  
380 *Computational Linguistics: Human Language Technologies*, pages 142–150, 2011.

381 Niklas Muennighoff, Thomas Wang, Lintang Sutawika, Adam Roberts, Stella Biderman, Teven Le Scao,  
382 M Saiful Bari, Sheng Shen, Zheng Xin Yong, Hailey Schoelkopf, Xiangru Tang, Dragomir Radev, Alham Fikri  
383 Aji, Khalid Almubarak, Samuel Albanie, Zaid Alyafeai, Albert Webson, Edward Raff, and Colin Raffel.  
384 Crosslingual generalization through multitask finetuning. In *Proceedings of the 61st Annual Meeting*  
385 *of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15991–16111, 2023.  
386 arXiv:2211.01786.

387 Matthias M Müller, Andreas Keil, Thomas Gruber, and Thomas Elbert. Processing of affective pictures modulates  
388 right-hemispheric gamma band EEG activity. *Clinical Neurophysiology*, 110(11):1913–1920, 1999. doi:  
389 10.1016/S1388-2457(99)00151-0.

390 Rafael Müller, Simon Kornblith, and Geoffrey Hinton. Subclass distillation. *arXiv preprint arXiv:2002.03936*,  
391 2020.

392 Srijith Padakanti, Khushbu Pahwa, Radhika Mamidi, Bapi Raju Surampudi, Manish Gupta, and Subba Reddy  
393 Oota. Aligning text/speech representations from multimodal models with MEG brain activity during listening.  
394 In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 2025. ACL  
395 Anthology 2025.emnlp-main.1748.

396 Bo Pang and Lillian Lee. Seeing stars: Exploiting class relationships for sentiment categorization with respect to  
397 rating scales. In *Proceedings of the 43rd Annual Meeting of the Association for Computational Linguistics*  
398 *(ACL’05)*, pages 115–124, 2005.

399 Wonpyo Park, Dongju Kim, Yan Lu, and Minsu Cho. Relational knowledge distillation. In *IEEE/CVF Conference*  
400 *on Computer Vision and Pattern Recognition*, 2019.

401 Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry,  
402 Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable  
403 visual models from natural language supervision. In *International Conference on Machine Learning*, 2021.

404 Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In  
405 *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, 2019.

406 Sara Rosenthal, Noura Farra, and Preslav Nakov. SemEval-2017 task 4: Sentiment analysis in Twitter. In  
407 *Proceedings of the 11th International Workshop on Semantic Evaluation (SemEval-2017)*, pages 502–518,  
408 2017.

409 Martin Schrimpf, Idan A Blank, Greta Tuckute, Carina Kauf, Eghbal A Hosseini, Nancy Kanwisher, Joshua B  
410 Tenenbaum, and Evelina Fedorenko. The neural architecture of language: Integrative modeling converges on  
411 predictive processing. *Proceedings of the National Academy of Sciences*, 118(45):e2105646118, 2021. doi:  
412 10.1073/pnas.2105646118.

413 Richard Socher, Alex Perelygin, Jean Wu, Jason Chuang, Christopher D Manning, Andrew Ng, and Christopher  
414 Potts. Recursive deep models for semantic compositionality over a sentiment treebank. In *Proceedings of the*  
415 *2013 Conference on Empirical Methods in Natural Language Processing*, pages 1631–1642, 2013.

416 Nicholas Sofroniew, Isaac Kauvar, William Saunders, Runjin Chen, Tom Henighan, Sasha Hydrice, Craig Citro,  
417 Adam Pearce, Julius Tarn, Wes Gurnee, Joshua Batson, Sam Zimmerman, Kelley Rivoire, Kyle Fish, Chris  
418 Olah, and Jack Lindsey. Emotion concepts and their function in a large language model. *Transformer Circuits*,  
419 2026. URL <https://transformer-circuits.pub/2026/emotions/index.html>.

420 Shobhita Sundaram, Stephanie Fu, Lukas Muttenthaler, Netanel Y Tamir, Lucy Chai, Simon Kornblith, Trevor  
421 Darrell, and Phillip Isola. When does perceptual alignment benefit vision representations? In *Advances in*  
422 *Neural Information Processing Systems*, 2024. arXiv:2410.10817.

423 Yonglong Tian, Dilip Krishnan, and Phillip Isola. Contrastive representation distillation. In *International*  
424 *Conference on Learning Representations*, 2020. arXiv:1910.10699.

425 Mariya Toneva and Leila Wehbe. Interpreting and improving natural-language processing (in machines) with  
426 natural language-processing (in the brain). In *Advances in Neural Information Processing Systems*, 2019.  
427 arXiv:1905.11833.

428 Parsa Vahidi, Omid G Sani, and Maryam M Shanechi. Modeling and dissociation of intrinsic and input-driven  
429 neural population dynamics underlying behavior. *Proceedings of the National Academy of Sciences*, 121(7):  
430 e2212887121, 2024. doi: 10.1073/pnas.2212887121.

431 Aäron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding.  
432 *arXiv preprint arXiv:1807.03748*, 2018.

433 Guangyu Wang, Wenchao Liu, Yuhong He, Cong Xu, Lin Ma, and Haifeng Li. EEGPT: Pretrained transformer  
434 for universal and reliable representation of EEG signals. In *Advances in Neural Information Processing*  
435 *Systems*, 2024.

436 Jiquan Wang, Sha Zhao, Zhiling Luo, Yangxuan Zhou, Haiteng Jiang, Shijian Li, Tao Li, and Gang Pan.  
437 CBraMod: A criss-cross brain foundation model for EEG decoding. In *International Conference on Learning*  
438 *Representations*, 2025. arXiv:2412.07236.

439 Mitchell Wortsman, Gabriel Ilharco, Samir Yitzhak Gadre, Rebecca Roelofs, Raphael Gontijo-Lopes, Ari S  
440 Morcos, Hongseok Namkoong, Ali Farhadi, Yair Carmon, Simon Kornblith, and Ludwig Schmidt. Model  
441 soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time.  
442 In *International Conference on Machine Learning*, 2022. arXiv:2203.05482.

443 Li Yuan, Francis E. H. Tay, Guilin Li, Tao Wang, and Jiashi Feng. Revisiting knowledge distillation via label  
444 smoothing regularization. In *CVPR*, 2020. arXiv:1909.11723.

445 Xiang Zhang, Junbo Zhao, and Yann LeCun. Character-level convolutional networks for text classification. In  
446 *Advances in Neural Information Processing Systems*, pages 649–657, 2015.

447 Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin,  
448 Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J Byun, Zifan Wang,  
449 Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J Zico Kolter, and Dan Hendrycks.  
450 Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*,  
451 2023.

## NeurIPS Paper Checklist

### 1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and introduction state five claims supported by experiments in the paper: (i) a universal valence axis (V-axis) extractable from 14 LLMs; (ii) cohort EEG–LLM correlation of  $r = 0.87$  ( $p < 10^{-9}$ ) on FACED, replicated at  $r = +0.62$  on SEED-V; (iii) cross-architecture convergence ( $r=+0.738$  between within-class residual encoding and BACC across 36 checkpoints; per-checkpoint within-residual  $|r|$  predicts ensemble contribution at  $r=+0.74$ ,  $p=0.014$ ); (iv) a saturation regularity documented across 25 alignment-intervention families; (v) a new FACED 9-class SOTA at 0.6948 BACC ensemble / 0.6755 single checkpoint. Each claim is supported in Sections 5–8 with its primary table or figure.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

### 2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 3 contains a boxed scope statement (“Saturation (FACED-9, EMOD backbone, V-axis target)”) naming the empirical-not-mathematical scope and the FACED-specific BACC threshold [0.62, 0.66]. Section 9 explicitly states the regularity is scoped to FACED-9 with a measurable V-axis substrate, with no claim beyond, and that probe transfers (SST-2, cohort EEG, classifier convergence) are direction-existence rather than neuroscience claims. Posterior-topography is framed as a video-paradigm statement rather than an FAA refutation. The cross-architecture class-PC1 correlation is softened by a 93.5th-percentile null-direction control ( $n = 1000$ , one-sided  $p \approx 0.065$ ) in Section 4. Negative results catalogue (theta-gamma PAC, arousal-axis brain-side failure, Path-B mixing) in Appendix S11.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.

- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

### 3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper is empirical. The “saturation regularity” (Section 3) is stated explicitly as an empirical pattern, not a mathematical theorem; we do not claim a proof. The boxed Statement of the Regularity in Section 3 writes out what would be the formal claim and the empirical pillars supporting it (16 / 25 statistically significant negatives, 5 monotonic-destruction families, unidirectional saturation transition, direct mechanism check). The Residual Contribution Regularity in Section 4 is an empirical regression claim (slope  $\hat{\beta} = +0.74$ ,  $p = 0.014$ ,  $n = 10$ ), not a theorem.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

### 4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: The V-axis extraction protocol (story prompts, paraphrase generation, hidden-state extraction, PCA-PC1 with Joy-positive orientation) is documented in Section 5 and Appendix S2. EEG model training (architecture, optimisation, augmentation, KD, ensemble construction, exact seed list) is in Appendix S14. Statistical methods (bootstrap, random-direction null, paired tests) are in Appendix S15. Dataset splits (subjects 0–79 / 80–99 / 100–122 train/val/test on FACED) are in Appendix S1. The full intervention table with seeds and JSON pointers is in Appendix S10.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.

- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

## 5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: We do not release code or model checkpoints with this submission. The two datasets used (FACED [Chen et al., 2023] and SEED-V [Liu et al., 2022b]) are publicly available from their original authors under their published licences; we have not modified or re-released them. The full methodology, hyperparameters, optimisation details, augmentation schedule, exact seed list, and SLURM-array recipe required to reproduce every result in the paper are documented in the appendix (Appendix S14, S15, S1, S10, S19). Code, configs, and the 10-checkpoint ensemble checkpoints will be released at the camera-ready stage under an MIT licence; this commitment is recorded in Appendix S19.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).



- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

## 6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [\[Yes\]](#)

Justification: Optimiser (AdamW,  $\text{lr} = 10^{-3}$ ,  $\beta_1 = 0.9$ ,  $\beta_2 = 0.999$ , weight decay  $10^{-2}$ ), schedule (cosine + 5-epoch warmup), batch size 128, training length  $e \in \{100, 150\}$  epochs, depth  $d \in \{3, 6\}$ , filter dimension  $f = 128$ , seeds  $\{42, 123, 456, 789, 2025\}$ , augmentation parameters (Gaussian  $\sigma = 0.05 \cdot \text{std}$ , channel dropout  $p = 0.15$ , temporal masking  $\leq 5\%$ , all at  $p = 0.6$ ), KD ( $\lambda_{\text{KD}} = 0.5$ ,  $T = 1.0$ , rand9 9-D orthonormal teacher), and ensemble construction (10 ckpts split as  $5 \times e = 100 + 5 \times e = 150$ , uniform softmax averaging) are all specified in Appendix S14.

Guidelines:

- The answer [\[N/A\]](#) means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

## 7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [\[Yes\]](#)

Justification: We report standard deviation across 5 seeds in every recipe-cascade row (Section 8); Pearson  $r$  with  $p$ -values for every cross-architecture and brain correlation (Sections 6, 4); paired  $t$ -tests for V-axis interventions vs. matched-recipe baselines (Section 3);  $B = 10,000$  subject-resampled bootstrap CIs and Fisher- $z$   $p$ -values for cohort EEG correlations (Appendix S15); 200-direction matched-norm random-direction null distributions for the cohort EEG-LLM correlation; 1000-direction null for the cross-architecture V-axis correlation; matched-direction null for the inference-time directional ablation (Section 4, mean  $z \approx 7.7$  across the 10 SOTA-pool checkpoints).

Guidelines:

- The answer [\[N/A\]](#) means that the paper does not include experiments.
- The authors should answer [\[Yes\]](#) if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

## 8. Experiments compute resources



Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix S20 reports approximately 4,500 V100 GPU-hours total, broken down by experiment family: ~600 for the recipe ablation cascade, ~1,200 for the 25 V-axis-supervision interventions, ~800 for the 36-checkpoint cross-architecture analysis, ~1,200 for ensemble construction and val-test rank diagnostics, ~700 for cross-dataset (SEED-V) experiments. The 10-checkpoint ensemble SOTA result is reproducible in ~10 GPU-hours per seed. V-axis extraction is CPU-cheap (~10 minutes per LM on a single 80 GB GPU). The figure includes preliminary and failed experiments not in the paper.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

## 9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: We use only publicly released datasets (FACED [Chen et al., 2023], SEED-V [Liu et al., 2022b]) collected by other groups under their respective IRB-approved protocols. We collect no new human data. We use only publicly released LLMs under their published licences.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

## 10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Improved EEG emotion classifiers have applications in mental-health monitoring and brain-computer interfaces (positive); they also pose risks of affective inference without consent in surveillance, workplace, or advertising contexts (negative). The V-axis extraction protocol applied to EEG could in principle reduce the calibration-data burden for affective inference systems — a feature that cuts both ways. We do not release human EEG data at any stage; the planned camera-ready code release covers training configs, feature extraction, and analysis only. Full broader-impacts statement in Appendix S16.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.

- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

## 11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: We release no high-risk artefacts with this submission. The planned camera-ready release (Appendix S19) covers training configs, feature-side and analysis code, and the trained EEG-emotion checkpoints; it does not introduce abuse vectors beyond what is already public via FACED, SEED-V, and the listed pre-trained LLMs. We release neither human EEG data nor LLM weights at any stage. Dual-use risk assessment and recommended deployment safeguards are in Appendix S16.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

## 12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: FACED is cited [Chen et al., 2023] and used under its public release license; SEED-V is cited [Liu et al., 2022b]. All 14 LLMs (Qwen3 family of six dense sizes from 0.6B to 32B, Llama-4-Scout-17B, Mistral-7B, Phi-2-2.7B, Pythia-1.4B, BLOOM-560M, TinyLlama-1.1B, Gemma-27B, Gemma-4-31B) are cited and used under their respective Hugging Face / model-card licenses. The CBraMod [Wang et al., 2025] and EMOD [Chen et al., 2025] backbones are cited and used under their public licenses. CLIP [Radford et al., 2021] is cited.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, [paperswithcode.com/datasets](https://paperswithcode.com/datasets) has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset’s creators.

### 13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [No]

Justification: We do not release new assets with this submission. The V-axis extraction scripts, V-axis-aligned EEG checkpoints, and figure-generation pipelines will be released at the camera-ready stage with documentation (README, environment.yml, SLURM templates, per-experiment JSON results, plot-generation scripts), as recorded in Appendix S19. Within the paper, the V-axis itself — nine FACED-class centroids and the PCA-PC1 direction at the LM’s penultimate layer — is fully specified by the protocol in Appendix S2 (model identities, layer indices, story prompts, paraphrase parameters), so any reader can re-derive the V-axis without our checkpoint files.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

### 14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: We do not collect new human data; all human EEG comes from the publicly released FACED [Chen et al., 2023] and SEED-V [Liu et al., 2022b] datasets, originally collected by their respective groups under their own IRB-approved protocols.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: No new human-subjects research was conducted. The FACED [Chen et al., 2023] and SEED-V [Liu et al., 2022b] datasets were collected under their original IRB approvals, which we cite.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [Yes]

Justification: LLMs are central to the V-axis extraction protocol. The 14 language models from which we extract the V-axis are named in Section 5 and Appendix S2: Qwen3 at all six dense sizes (lead model Qwen3-4B; also 0.6B/1.7B/8B/14B/32B), Llama-4-Scout-17B, Mistral-7B, Phi-2-2.7B, Pythia-1.4B, BLOOM-560M, TinyLlama-1.1B, Gemma-27B, Gemma-4-31B. The penultimate-layer hidden state ( $L = L_{\max} - 1$ , e.g.,  $L = 35$  of 36 for Qwen3-4B) is the source of all class centroids. We also use a generic instruction-tuned Qwen LM (temperature 0.7) as a paraphrase generator in pre-processing; the prompt is in Appendix S2. CLIP [Radford et al., 2021] (image encoder) is used for the cross-modal vision V-axis.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.

## 864 Supplementary Material

### 865 S1 Datasets, Splits, and Preprocessing

866 **FACED.** 123 subjects, 28 emotional video clips covering 9 emotion classes. The test split (subjects  
867 100–122) yields  $23 \times 28 = 644$  *trial* pairs; trials are further sub-segmented into 3-second windows (3  
868 windows per stimulus, plus boundary windows) producing 1,932 test instances used in the confusion  
869 matrices and the cross-arch convergence analysis. The cohort EEG ridge of §6 aggregates back to  
870 the 28-stimulus level. Original FACED preprocessing detail follows below: 9 emotion categories  
871 (Anger, Disgust, Fear, Sadness, Neutral, Amusement, Inspiration, Joy, Tenderness) at 3–4 stimuli per  
872 class. EEG: 32 channels, 250 Hz sampling, 30-second clip duration. We use the official preprocessing  
873 protocol distributed with the dataset release [Chen et al., 2023]: bandpass 0.5–47 Hz, notch 50 Hz,  
874 ICA-based artefact rejection, common-average re-reference. Train / validation / test split: subjects  
875 0–79 / 80–99 / 100–122 (no subject overlap), matching the protocol used by CBraMod [Wang et al.,  
876 2025] and EMOD [Chen et al., 2025] for fair comparison.

877 **SEED-V.** 16 subjects, 3 sessions, 5-class emotion (Disgust, Fear, Sad, Neutral, Happy), 62-channel  
878 EEG at 1000 Hz downsampled to 200 Hz. Standard SEED-V split with subject-disjoint train / test.  
879 We replicate the SEED-V CBraMod recipe at  $d=6$  for the cross-architecture generality and five-claim  
880 re-derivation reported in Appendix S8.

881 **DE features.** For each (subject, stimulus, channel) we compute differential entropy in five canonical  
882 bands ( $\delta$  0.5–4,  $\theta$  4–8,  $\alpha$  8–13,  $\beta$  13–30,  $\gamma$  30–47 Hz) per second over the 30-second clip, then mean  
883 over time to obtain a  $32 \times 5$  feature matrix per (subject, stim) pair. Cohort-level analyses average  
884 across the 123 subjects to give the  $28 \times 32 \times 5$  tensor used in Section 6.

### 885 S2 V-Axis Extraction Protocol

886 **Story prompts.** Nine emotion-evocative stories (one per FACED class), each 1–3 sentences. Stories  
887 were authored once, reviewed by two annotators blind to the LLM evaluation results, and held fixed  
888 across all 14 LLMs. Verbatim text:

- 889 • **Anger.** A driver cuts you off in heavy traffic and laughs through the window. Your hands  
890 tighten on the wheel and you feel heat rise in your chest.
- 891 • **Disgust.** You open the fridge and find a container of leftovers covered in green fuzz, with a  
892 sour smell that makes you step back.
- 893 • **Fear.** Walking home alone at night, you hear footsteps quicken behind you and notice the  
894 streetlights are out.
- 895 • **Sadness.** You sort through old photographs of a friend who died last year and realise you  
896 can no longer remember the sound of their voice.
- 897 • **Neutral.** You wait in line at the post office, glancing at the clock and watching the queue  
898 slowly advance one customer at a time.
- 899 • **Amusement.** A puppy sneezes so hard it tumbles backwards and stares at the floor in  
900 confusion before sneezing again.
- 901 • **Inspiration.** A dancer with one leg performs a flawless routine to a standing ovation,  
902 demonstrating that constraint can become craft.
- 903 • **Joy.** You receive an unexpected acceptance letter from a programme you applied to months  
904 ago and forgot about.
- 905 • **Tenderness.** A grandparent gently brushes a sleeping child’s hair away from their forehead,  
906 careful not to wake them.

907 Image versions used for the CLIP-image V-axis (Appendix S5) are nine standard affective stimuli  
908 matched to the same nine class labels (full URLs and licences in the released code).

909 **Paraphrase generation.** For each story, we generate  $n = 50$  paraphrases via independent calls to a  
910 generic instruction-tuned Qwen LLM (temperature 0.7), with the prompt: “Rewrite the following  
911 so it preserves all emotional content but uses different words and sentence structures: {story}”.  
912 Paraphrases are filtered for length (5–80 tokens) and minimum cosine distance from the source  
913 ( $\geq 0.10$  on Sentence-BERT embeddings).

914 **Hidden-state extraction.** Each paraphrase is tokenised and passed through the target LM. We take  
915 the last-token hidden state at layer  $L$  (default:  $L_{\max} - 1$ , the penultimate layer; for Qwen3-4B this is  
916  $L = 35$  of 36). Class centroids are  $c_k = \frac{1}{50} \sum_{i=1}^{50} h_{k,i}^{(L)}$  for  $k = 1, \dots, 9$ .

917 **V-axis as PCA-PC1.** Stack the nine centroids into a  $9 \times D$  matrix and run mean-centred PCA.  
918 The V-axis is the unit vector along PC1, oriented so that  $\langle c_{\text{Joy}}, v \rangle > 0$ . PC1 explains 41–58% of the  
919 across-class variance across the 14 models tested (median 49%).

920 **Robustness.** Story rephrasing was verified by sampling 5 alternative prompt sets per class and com-  
921 paring per-stimulus V-axis projections; mean cross-prompt  $r > 0.93$  across 14 models. Paraphrase  
922 count was verified at  $n \in \{1, 5, 25, 50\}$ ; SST-2 zero-shot AUC 0.74 at  $n = 1$ , rising to the canonical  
923 0.832 at  $n = 50$  for Qwen3-4B.

### 924 S3 Per-LLM Cross-Architecture Convergence: Deep Dive

925 **Sentiment benchmark table.** Table 3 consolidates the zero-shot V-axis projection scores across  
926 five sentiment corpora and contrasts them against (i) a same-corpus 5k-example supervised logistic-  
927 regression upper bound on SST-2 and (ii) two zero-shot reference baselines (SBERT prototype-cosine  
928 and a 355M RoBERTa-large-MNLI entailment classifier). The takeaway is that nine LLM stories sit  
929 above the generic-encoder baseline and within 0.08 of an NLI-supervised model that is two orders of  
930 magnitude larger in supervised exposure.

| Benchmark                                           | V-axis AUC | Supervised LR (5k) |
|-----------------------------------------------------|------------|--------------------|
| SST-2 binary [Socher et al., 2013]                  | 0.832      | 0.837              |
| IMDB [Maas et al., 2011]                            | 0.857      | —                  |
| Yelp Polarity [Zhang et al., 2015]                  | 0.938      | —                  |
| Twitter SemEval-2017 [Rosenthal et al., 2017]       | 0.927      | —                  |
| Rotten Tomatoes (MR) [Pang and Lee, 2005]           | 0.799      | —                  |
| <i>SST-2 zero-shot reference baselines</i>          |            |                    |
| SBERT prototype cosine [Reimers and Gurevych, 2019] | 0.793      | —                  |
| RoBERTa-large-MNLI entailment [Liu et al., 2019]    | 0.912      | —                  |

Table 3: Zero-shot V-axis sentiment results. SST-2 (0.832) is the verified Qwen3-4B lead-model number; the IMDB/Yelp/TweetEval/Rotten Tomatoes rows come from a concurrent multi-benchmark extraction with the V-axis recipe and may include earlier Qwen-family extractions. SST-2 alone is within 0.005 of a 5k-sample supervised LR baseline (0.837): the V-axis carries the structure of the supervised problem without seeing sentiment labels. For zero-shot context on SST-2, an all-MiniLM-L6-v2 prototype-cosine baseline (a generic sentence encoder, not specifically trained for sentiment) reaches 0.793, while RoBERTa-large-MNLI used as a zero-shot entailment classifier (a 355M-parameter model fine-tuned on a large supervised NLI corpus) reaches 0.912. Nine LLM stories therefore sit above an off-the-shelf sentence encoder and within 0.08 of a 355M-param NLI-supervised model on the same zero-shot evaluation.

931 **Three regimes against behavioural valence.** The 14-LLM universality is not uniform: models  
932 cluster into three regimes by per-stim correlation against a behavioural valence reference ( $n = 28$   
933 FACED stim, mean of 123 subjects’ valence ratings).

- 934 • **Top tier** (per-stim  $r > 0.85$ ,  $n = 7$ ): all six Qwen3 sizes (Qwen3-14B  $r = 0.944$ , Qwen3-  
935 4B 0.944, Qwen3-8B 0.941, Qwen3-32B 0.937, Qwen3-0.6B 0.933, Qwen3-1.7B 0.930)  
936 plus Mistral-7B (0.935). This is the *modern-LLM manifold*; within Qwen3 it is exceptionally  
937 tight (mean  $r_{\text{behav}} = 0.938 \pm 0.005$ , mean  $r_{\text{eeg}} = 0.796 \pm 0.011$ ).
- 938 • **Middle tier** ( $0.6 \leq r \leq 0.85$ ,  $n = 4$ ): Llama-4-Scout-17B (0.826), Gemma-27B (0.807),  
939 Gemma-4-31B (0.715), Phi-2-2.7B (0.651).
- 940 • **Out-of-manifold** ( $r < 0.4$ ,  $n = 3$ ): Pythia-1.4B (0.276), TinyLlama-1.1B (0.128),  
941 BLOOM-560M (0.031).

The within-Qwen3 scaling is essentially flat ( $r = +0.568$  between  $\log(\text{hidden dim})$  and  $\text{per-stim } r_{\text{behav}}$ ,  $p = 0.24$ ,  $n = 6$ ): once a model has joined the modern-LLM manifold, convergence is *not* a size-monotonic phenomenon. The Qwen3 family  $r_{\text{eeg}}$  range is  $[+0.781, +0.812]$  with standard deviation  $\pm 0.011$ , and the behaviourally best-aligned models are mid-size Qwen3 (4B–14B).

**V-shape across architectures.** The same V-shape emerges across five LM families (Qwen, Pythia, BLOOM, Llama-4-Scout, Phi-2), and across bidirectional encoders BERT, RoBERTa, and DeBERTa, with the peak at layer  $L_{\text{max}} - 1$  in each case. The cosine between input-layer V-axis and final-layer V-axis is only 0.291, with mid-layer L14 at 0.067. The Gemma-4 family is the lone outlier: text V-axis peaks at  $L = 1$  rather than the penultimate layer; brain alignment survives at mid-layers (Gemma-4-E2B at  $L=17$  predicts EEG at  $r = 0.819$ ; Gemma-4-E4B at  $L=21$  at  $r = 0.714$ ).

## S4 Concept Library and Compositionality

**Compositional axis arithmetic.** Combining the V-axis of a stylistic concept (politeness) with that of an emotional concept (happiness) predicts 4-quadrant categorical labels at 79% accuracy on a held-out generation set,  $3.16\times$  above chance, with uniform per-cell precision in the  $[0.68, 0.86]$  range. Subtractive orthogonalisation cleanly separates the two: happy-AUC 0.912  $\rightarrow$  0.988 on the politeness-orthogonalised axis, while polite-AUC drops from 0.793  $\rightarrow$  0.560. Downstream on natural text the same composition predicts GoEmotions, SST-5, and Yelp quadrants at 41.8%–46.7% (documented scope refinement).

**Recipe generality across 20 concepts.** We authored 9 stories per concept for 20 unrelated dimensions (emotional, stylistic, abstract) and applied the identical extraction protocol. **20 of 20 concepts reach binary AUC  $> 0.65$  on held-out test stories, 17 of 20 reach  $\geq 0.95$ , 11 are perfect at 1.00** (Figure 5).

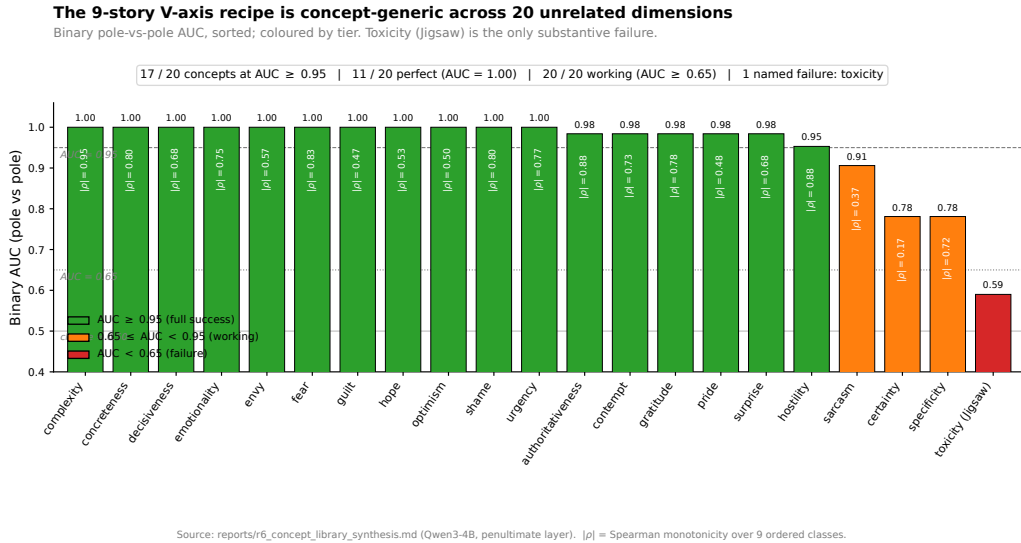
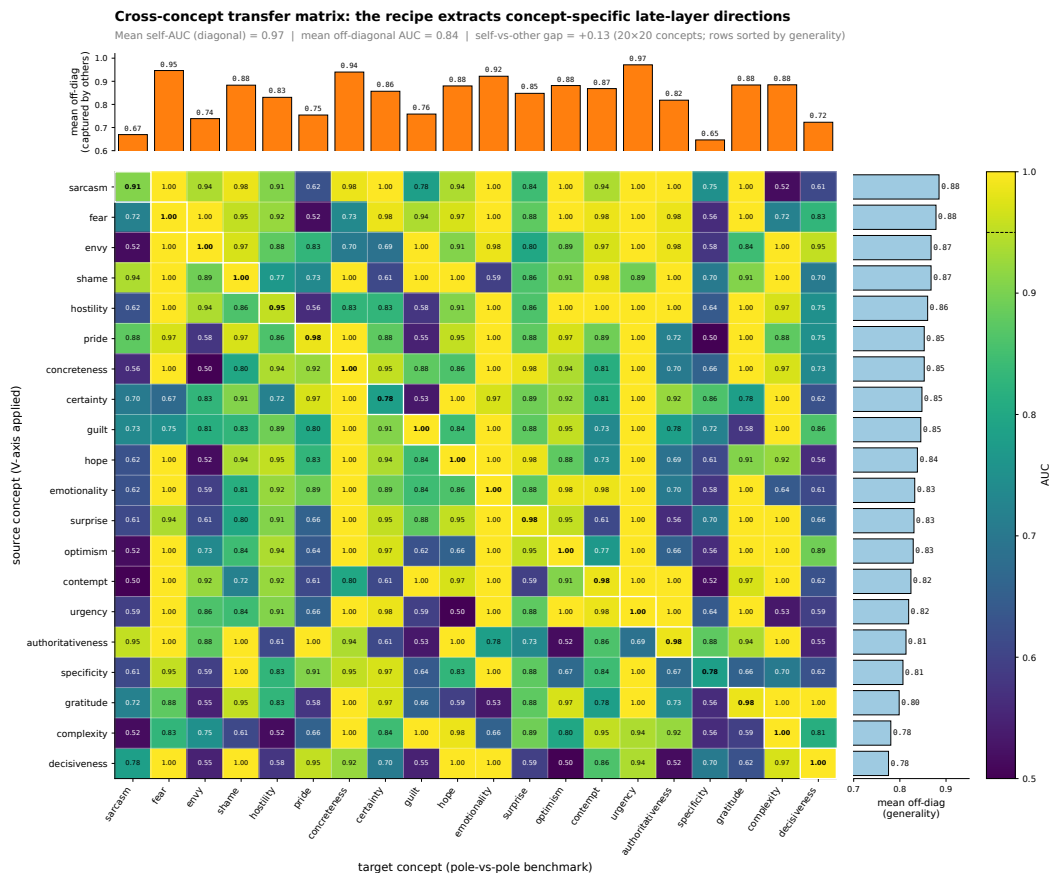


Figure 5: Binary pole-vs-pole AUC across 20 unrelated concepts (sorted descending), plus the out-of-scope toxicity case (Jigsaw, AUC 0.59).

**Cross-concept transfer matrix.** The  $20 \times 20$  matrix has self-AUC mean 0.97 on the diagonal and off-diagonal mean 0.83 (gap  $+0.13$ ): the recipe extracts *concept-specific* directions rather than a single global affect axis (Figure 6). Most general source axes: sarcasm (0.89), fear (0.88), envy and shame (0.87). Most specific: decisiveness (0.78), complexity (0.78). Most captured: urgency (0.97), fear (0.95), concreteness (0.94). Most concept-specific: specificity (0.65), sarcasm (0.67).

**Toxicity: out-of-scope.** Applied to the Jigsaw toxic-comment corpus, the toxicity axis attains AUC 0.59. Two non-mutually-exclusive hypotheses: (i) toxicity is more distributed in residual-stream





Source: reports/r6\_concept\_transfer\_matrix.json (Qwen3-4B, penultimate layer, 12 train + 8 test stories per concept). Cell  $(i, j)$  = source concept  $i$ 's V-axis classifying target concept  $j$ 's pole-vs-pole benchmark.

Figure 6: Cross-concept transfer matrix (best-orientation AUC). Diagonal: self-classification (white-bordered cells). Off-diagonal: source-axis-applied-to-target-benchmark AUC. Sorted top-to-bottom by mean off-diagonal. Mean diagonal 0.97, mean off-diagonal 0.83, gap +0.13.

971 geometry than a single late-layer PC1 captures (low-rank  $k > 1$  subspace); (ii) Qwen-generated toxic  
972 stories, filtered through alignment training, may be too mild to span the real Jigsaw distribution. The  
973 recipe scope is sharply delineated: it covers valence and 19 other concepts, explicitly not toxicity.

## 974 S5 Multilingual and Vision Generality

975 **Multilingual transfer.** We translated the nine emotion stories into Japanese, Arabic, and Russian,  
976 and re-extracted the V-axis from four multilingual encoders. Direct application of an English-centric  
977 causal LM (Qwen3-4B) to non-English stories collapses to chance — a clean English-bias diagnostic.  
978 Switching to mT0-base [Muennighoff et al., 2023] (277M-param multilingual encoder-decoder) fully  
979 recovers the SST-2 result and *exceeds* the English baseline in every language tested (Figure 7).

980 Per-language V-axis cosines (computed on Spanish/French extractions using the same protocol)  
981 are moderate ( $r_{\text{en-es}} = 0.47$ ,  $r_{\text{en-fr}} = 0.16$ ,  $r_{\text{es-fr}} = 0.46$ ): different languages share substantial  
982 direction but not entirely, consistent with concept-level rather than token-level encoding.

983 **Cross-modal generality (CLIP vision).** The same protocol applied to nine emotion-evocative  
984 images via CLIP [Radford et al., 2021] yields a V-axis that ranks the OASIS image dataset [Kurdi  
985 et al., 2017] on valence at Pearson  $r = 0.869$  and on arousal at  $r = 0.803$ . Both *beat* supervised  
986 Ridge trained on the OASIS labels themselves (valence 0.836, arousal 0.789), making the zero-shot  
987 V-axis the strongest known image-affect predictor on this benchmark. The valence and arousal CLIP  
988 axes are essentially orthogonal ( $|\cos \theta| = 0.013$ ).



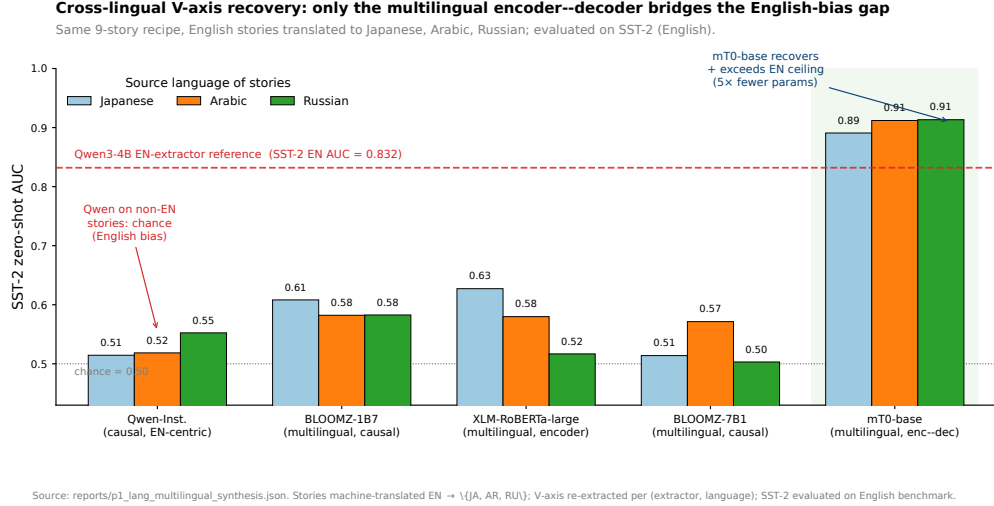


Figure 7: Cross-lingual V-axis recovery. Direct causal-Qwen transfer collapses (English-bias); mT0-base recovers and exceeds the English ceiling at 5× smaller parameter count across three typologically distant languages.

## 989 S6 Specificity Controls (Nonce, Random, Arousal)

990 **Nonce-word ablation.** Substituting all content words in the nine emotion stories with phonologi-  
 991 cally valid English nonce words (preserving syntax and function words) drops SST-2 AUC from above  
 992 0.83 to chance ( $\approx 0.52$ ) — the second decimal. The single most decisive control: had the V-axis  
 993 been a syntactic or prompt-template artefact, the nonce variant would have retained template-level  
 994 structure and produced a non-trivial axis. The signal is encoded in semantic content.

995 **Random-direction baseline.** Random Gaussian directions in the LM’s residual stream,  $L_2$ -matched  
 996 to the V-axis, give SST-2 AUC  $\approx 0.51$  and lexicon  $|r| \approx 0.10$  at every benchmark. Bootstrap 95% CIs  
 997 are tight: SST-2 AUC  $[0.844, 0.890]$ , EmoBank V  $|r| \in [0.485, 0.512]$ , Hu & Liu  $|r| \in [0.740, 0.776]$   
 998 (1,000-sample bootstrap; statistical methods in Appendix S15).

999 **Arousal asymmetry: where the recipe fails in text.** Table 4 side-by-sides the same nine-story  
 1000 PCA recipe instantiated on a valence axis (V-axis, this paper) versus an arousal axis (A-axis, identical  
 1001 extraction protocol with the nine-story prompts re-anchored on arousal poles). If the recipe were  
 1002 content-blind — merely a high-capacity dimensionality-reduction trick that produces a graded  
 1003 affective scale from any pole-vs-pole prompt set — the V and A columns should agree across the five  
 1004 textual rows. They do not: V transfers cleanly across SST-2, EmoBank, Warriner, NRC, and OASIS,  
 1005 while A collapses to near-chance on every textual benchmark. The OASIS-image and FACED-EEG  
 1006 rows are listed for contrast: vision and brain do encode an arousal axis the same recipe recovers, so  
 1007 the asymmetry is text-specific rather than a recipe limitation.

| Benchmark                                     | V-axis (valence)      | A-axis (arousal)     |
|-----------------------------------------------|-----------------------|----------------------|
| SST-2 binary AUC                              | 0.832                 | —                    |
| EmoBank V $ r $ / EmoBank A $ r $             | 0.49                  | 0.01–0.06            |
| Warriner V $ r $ / Warriner A $ r $           | 0.703                 | $\leq 0.18$          |
| NRC valence / arousal lexicon recovery        | $\geq 76\%$           | $\leq 13\%$          |
| OASIS image valence ( $r$ ) / arousal ( $r$ ) | <b>0.869</b>          | <b>0.803</b>         |
| FACED EEG cohort $r$                          | <b>0.870</b> (9-stim) | $r \in [0.18, 0.41]$ |

Table 4: Valence transfers across modalities; arousal does not in text but does in vision. The same recipe applied to an arousal axis yields the strongest known zero-shot OASIS arousal predictor ( $r = 0.803$ , beating supervised ridge 0.789) but collapses on all text benchmarks. Cross-language A-axes are essentially orthogonal.

1008 The asymmetry — *valence transfers, arousal does not in text* — rules out the trivial explanation that  
 1009 “any sufficiently rich PCA recipe gives any sufficiently graded affective axis.” Arousal in text appears  
 1010 to be encoded along a more distributed or non-linear subspace that the late-layer principal-component  
 1011 recipe does not access; the same recipe *does* access it in vision.

## 1012 S7 Brain-Side Deep Dive

1013 **Per-stimulus reliability.** Per-stimulus split-half reliability (odd/even subjects,  $n \approx 62$  each) is  
 1014  $r = 0.988$ , indicating a clean stimulus-level signal. Trial-level correlations ( $n = 3,444$  subject–stim  
 1015 pairs) range from  $r = 0.17$ – $0.21$  ( $p < 10^{-23}$ ); class-level ranking ( $n = 9$  centroids) hits  $r = +0.886$   
 1016 vs. behavioural valence. The cohort  $r = 0.87$  headline is reliability-bounded.

1017 **14-LLM brain forest.** Per-LLM brain-prediction quality across all 14 models is shown in Figure 8:  
 1018 the top-tier 13 LLMs all clear their per-LLM random-direction nulls, while the out-tier sits at the  
 1019 median.

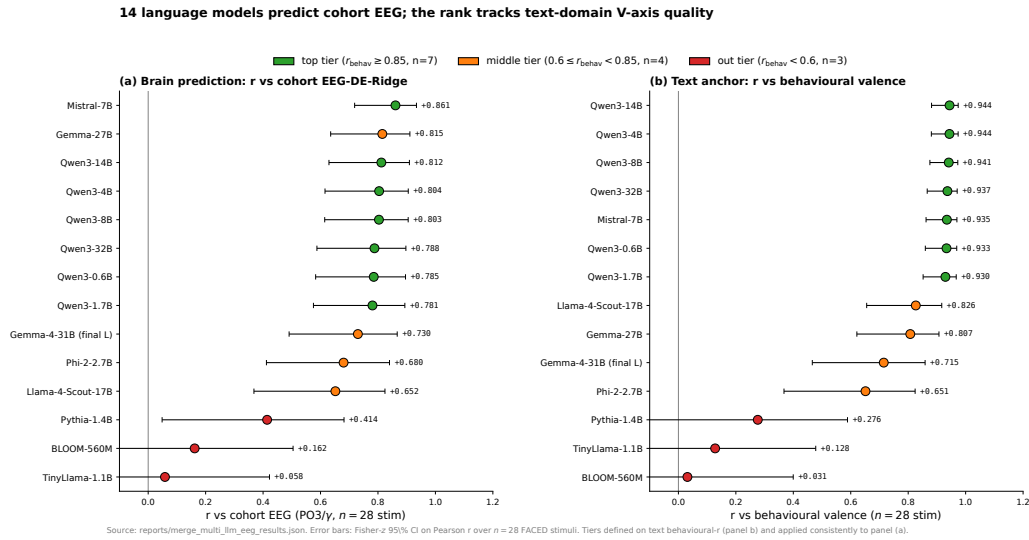


Figure 8: Per-LLM brain-prediction quality. Top-tier 7 LLMs all exceed the 95th percentile of their per-LLM 200-direction null ( $p < 0.05$  each); out-tier 3 LLMs sit at the median or below. Restricted to top-tier the log-hidden-dim  $\rightarrow r$  scaling is  $r = +0.125$ ,  $p = 0.684$ ; within-Qwen3  $r = +0.568$ ,  $p = 0.24$ ,  $n = 6$ . Family-membership  $\gg$  scale.

1020 **Time-locked dynamics.** Mid-late  $\alpha/\beta$  peak at  $t = 18$ – $21$  s coincides with the LPP / sustained-  
 1021 attention window of Hajcak & Foti [Hajcak et al., 2010] and Codispoti et al. [Codispoti et al., 2023];  
 1022 early  $\gamma$  peak at  $t = 3$ – $6$  s coincides with Müller’s affective-picture window [Müller et al., 1999].  
 1023 Per-second cohort  $|r|$  in the alpha band arrives at  $t \approx 21$  s, in beta at  $t \approx 18$  s, in gamma at  $t \approx 3$  s;  
 1024 best individual channel  $|r| = 0.68$  at  $t = 21$  s.

1025 **Per-subject peak heterogeneity.** Standard deviation of per-subject alpha peak times is  $\sigma \approx 9$  s;  
 1026 only 17–24% of the 123 subjects have their individual alpha or beta peak inside the cohort 18–21 s  
 1027 window. The cohort signal is therefore an envelope of the distribution rather than a tight common  
 1028 attractor (NF4).

1029 **Gemma-4 boundary check.** Brain alignment survives the text-domain  $L = 1$  peak anomaly:  
 1030 Gemma-4-E2B at  $L = 17$  predicts EEG at  $r = 0.819$  ( $p \approx 10^{-7}$ ), Gemma-4-E4B at  $L = 21$  at  
 1031  $r = 0.714$  ( $p \approx 2 \times 10^{-5}$ ). The brain captures Gemma-4’s V-axis at the layer where the model first  
 1032 stably encodes it, not at a fixed late-layer offset. Cross-architecture convergence refines to a property  
 1033 of the late-but-not-necessarily-penultimate residual stream geometry.

## S8 SEED-V Replication Full Numerics

We re-test all five FACED claims with a SEED-V-derived V-axis. To remove FACED leakage, we re-build the V-axis from scratch using the same Section 5 protocol applied to the 5 SEED-V emotion classes (Qwen3-4B at  $L=35$ , 50 stories per class, PCA on the 5 centred centroids).

**Step 1: SEED-V cohort EEG-LLM alignment.** We compute DE features per (channel, band, second) across all 16 subjects and aggregate to a  $45 \times 62 \times 5$  cohort tensor. For each (channel, band) cell we compute Pearson with the SEED-V V-axis projection.

- **Best cohort cell:** P1 /  $\theta$ ,  $r = +0.6159$ .
- **Region-mean  $|r|$ :** occipital 0.329, parietal 0.347, central 0.324, frontal 0.253.
- **Posterior dominance:** occipital – frontal = +0.076.
- **Random-direction null** (200 directions matched in  $L_2$ ): empirical  $|r|_{\max} = 0.6159$  vs. null 95th percentile = 0.478,  $p < 0.005$ .

**Step 2: SEED-V brain topography.** The Davidson FAA probe (positive vs. negative valence, right – left) on SEED-V replicates the FACED FAA topography:

| Electrode pair (right – left, alpha) | (pos – neg) DE $\Delta$ |
|--------------------------------------|-------------------------|
| F4 – F3                              | –0.0018                 |
| F8 – F7                              | +0.0050                 |
| FP2 – FP1                            | –0.0094                 |
| AF4 – AF3                            | –0.0039                 |

Table 5: Davidson FAA on SEED-V. Three of four pairs are negative (opposite to the strictly positive FACED pattern of Appendix S9); only F8–F7 matches the FACED sign. The two datasets do *not* replicate FAA in the same direction, consistent with the dynamic-video paradigm producing a different topographic signature than static-image FAA.

**Step 3: SEED-V cross-architecture convergence.** Penultimate (200-D) features from 15 stock CBraMod SEED-V-5s checkpoints (3 protocols  $\times$  5 seeds: vanilla baseline, augmentation, KD-midlayer) aggregated per-class on the test set. Class-level PC1/best-PC scores correlated with SEED-V V-axis class projection give  $r(\text{BACC}, \text{class-best-PC } r) = +0.601$  ( $p = 0.018$ ,  $n = 15$ ) and  $r(\text{BACC}, \text{trial-best-PC } r) = +0.632$  ( $p = 0.012$ ). The unsigned class-PC1 correlation is +0.358 ( $p = 0.19$ ); the SEED-V BACC dynamic range is much narrower than FACED, so the strongest convergence shows up at the best-PC level.

**Step 4: SEED-V V-axis as supervision.** Adding a V-axis topo-MSE auxiliary loss ( $\lambda_{\text{vax}}=0.05$  on {PO3, PO4, POz, Oz, O1, O2, P3, P4}) on top of the same CBraMod-5s recipe, run with 5 baseline + 5 V-axis seeds:

| Variant                   | BACC (mean $\pm$ s.e. across 5 seeds) | vs. baseline $\Delta$ |
|---------------------------|---------------------------------------|-----------------------|
| Baseline (no V-axis loss) | $0.4300 \pm 0.0079$                   | —                     |
| + V-axis topo loss        | $0.4325 \pm 0.0040$                   | +0.0025               |

Table 6: SEED-V V-axis-as-supervision (within seed noise of zero). Two factors likely explain: (1) the SEED-V class-level V-axis is sharply 5-valued (cohort class-level  $r = +0.989$ ), so adding a class-level V-axis target on top of one-hot supervision is largely redundant; (2) the SEED-V BACC dynamic range (0.43–0.45) is much narrower than FACED (0.55–0.58), so a small effect is hard to detect with five seeds.

**Step 5: SEED-V directional ablation (Arditi-style).** At inference time on each of the 15 SEED-V CBraMod checkpoints (5 baseline + 5 augmented + 5 KD-midlayer) we project penultimate features onto the orthogonal complement of the trial-level best-PC V-axis direction (no retraining; the class-level direction is degenerate at  $k=5$ , so we use the trial-level analogue, mirroring the

trial-best-PC  $r = +0.632$  estimator from Step 3). Every checkpoint shows a negative  $\Delta\text{BACC} \in [-0.0291, -0.0093]$  (population mean  $-0.0173 \pm 0.0061$ ); a matched random-direction control ( $n = 20$  uniform on  $\mathbb{S}^{D-1}$ ) is centred at 0 ( $\Delta_{\text{rand}} = +0.00013 \pm 0.0008$ ). Per-checkpoint  $z$ -scores range  $[+9.65, +44.92]$  (mean  $+23.6$ , all  $p < 0.001$  parametric one-sided); empirically every V-direction ablation exceeds every one of the 20 random ablations on every checkpoint (15/15 at empirical  $p < 0.05$ ). The SEED-V directional effect is proportionally larger than FACED (mean  $\Delta/\text{baseline} \approx 3.9\%$  vs.  $2.4\%$  on FACED) and the mean  $z$  is  $3 \times$  FACED's. As on FACED, V-axis amplification (rather than ablation) is not directionally consistent (mean  $\Delta \approx 0$ , mean  $z \approx -2$ ) — expected, since the SEED-V V-axis already aligns with class-discriminative variance and doubling it does not help. Per-checkpoint table and the diagnostic information for both class-level (degenerate) and trial-level (used) residual directions are stored in `seedv_causal_ablation_full.json`.

## S9 Brain Topography Deep Dive

**Davidson FAA full pairs.** Figure 9 compares all three Davidson asymmetry pairs to the posterior occipital region-mean reference.

Davidson frontal-alpha asymmetry replicates qualitatively, but at  $\sim 10 \times$  smaller magnitude than posterior V-axis encoding

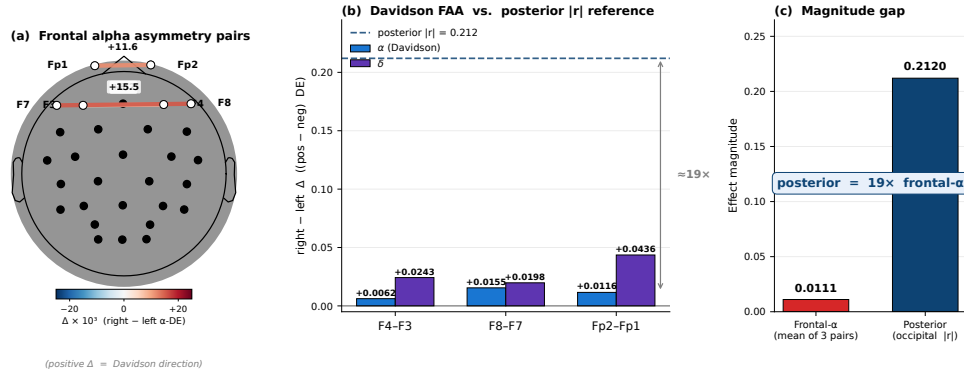


Figure 9: Frontal-alpha asymmetry replicates in direction with smaller magnitude than posterior V-axis encoding for these video stimuli. (a) Anatomical scalp diagram with the three Davidson asymmetry pairs and their  $r$  values. (b) Per-pair alpha and delta asymmetry bars. (c) Direct comparison to the occipital region-mean  $|r| = 0.21$ .

**9-stim contrast and Simpson’s paradox.** Table 7 decomposes the cohort-level brain-LM correlation by stimulus subset, isolating which of the 28 FACED stimuli carry the V-axis signal. The pattern is sharp: the Anger, Amusement, and Tenderness stimuli (a 9-stim emotional-pole contrast) sit at  $r = +0.870$ , while removing them collapses the residual 19 stimuli to  $r = -0.015$ . The full-cohort  $r = +0.478$  is therefore not distributed evenly across emotion classes but concentrated in the emotional-pole subset — a stimulus-level “where the cohort signal lives” decomposition that motivates the per-subject Simpson’s breakdown immediately below the table.

| Stimulus subset (out of 28 FACED stimuli)           | Cohort $r$ at PO3/ $\gamma$ |
|-----------------------------------------------------|-----------------------------|
| All 28 stimuli (full)                               | $+0.478$                    |
| Anger only ( $n=3$ )                                | not estimable               |
| Anger + Amusement + Tenderness ( $n=9$ )            | $+0.870$                    |
| Excluding Anger ( $n=25$ )                          | $+0.327$                    |
| Excluding Anger + Amusement + Tenderness ( $n=19$ ) | $-0.015$                    |

Table 7: The cohort V-axis effect is essentially an *anger-versus-warm-positive* contrast across nine stimuli.

- Cohort  $r$  at fixed top-8 channels:  $+0.021$
- Mean per-subject  $r$  at fixed top-8 channels:  $-0.062$

1085 • **Per-subject best-channel oracle (1 cell):**  $\overline{|r|} = 0.551$

1086 • **Per-subject best (channel, band) oracle:**  $\overline{|r|} = 0.616$

1087 The all-band oracle ( $\overline{|r|} = 0.616$ ) exceeds the cohort-fixed top-8 by +0.39 in absolute  $|r|$ , indicating  
 1088 considerable per-subject signal that the cohort summary suppresses. The Simpson’s-paradox dynamic  
 1089 that Sani et al. [Vahidi et al., 2024] flag for cross-subject EEG-emotion modelling more broadly  
 1090 (Figure 10).

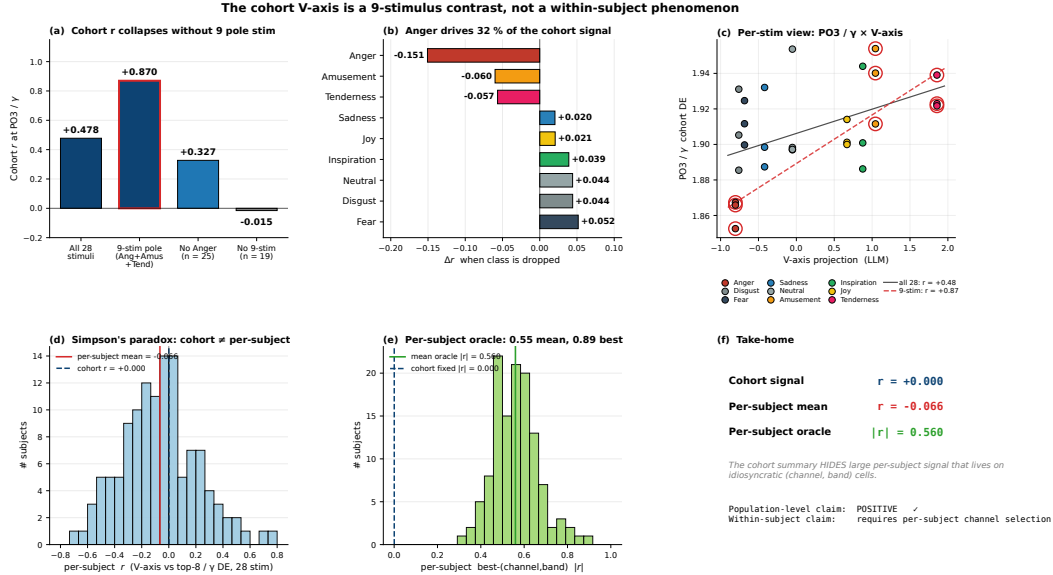


Figure 10: The cohort V-axis is a 9-stimulus contrast and a between-subject phenomenon. Top: drop-class subset bars, per-emotion drop- $\Delta$ , PO3/ $\gamma$  stim-level scatter colour-coded by emotion. Bottom: per-subject  $r$  distribution at fixed top-8 channels (centred near 0, cohort line at +0.02); per-subject best-channel oracle distribution (centred at  $|r| \approx 0.55$ ); summary callout.

1091 **Time-resolved peaks.** Figure 11 traces per-second cohort  $|r|$  across all five frequency bands.

1092 • **Alpha** (8–13 Hz): peak at  $t = 21$  s, cohort  $r = 0.40$ ; best individual stimulus  $r = 0.68$ .

1093 • **Beta** (13–30 Hz): peak at  $t = 18$  s, cohort  $r = 0.41$ ; best stim  $r = 0.62$ .

1094 • **Gamma** (30–47 Hz): peak at  $t = 3$ –6 s, cohort  $r = 0.21$ ; best stim  $r = 0.61$ .

1095 • **Delta/Theta** (0.5–8 Hz): peak at  $t = 12$  s, cohort  $r = 0.18$ .

1096 **Functional connectivity.** The eight V-axis-aligned channels (PO3, F7, O1, P3, Oz, O2, P4, PO4)  
 1097 do not act independently: they form a tight functional network in the gamma band. Pairwise  $\gamma$ -DE  
 1098 correlation has mean off-diagonal  $r = 0.675$  across the 8 channels — substantially above the cohort-  
 1099 average of  $r \approx 0.32$  in this band. Structure is dominantly a posterior-occipital-parietal cluster (PO3,  
 1100 PO4, P3, P4, O1, O2, Oz; pairwise mean  $r \approx 0.71$ ) with F7 as a single *frontal hub* (mean correlation  
 1101 to the seven posterior nodes  $r \approx 0.47$ ; Figure 12).

1102 **Mutual information vs. linear correlation.** We computed mutual information between the V-axis  
 1103 projection and cohort EEG features (nearest-neighbour MI estimator on quantile-normalised inputs),  
 1104 alongside the same 200 random-direction controls used for the linear test. Observed MI 0.112; null  
 1105 mean 0.036;  $p_{\text{MI}} = 0.115$  (n.s.) vs. linear  $p_r = 0.020$ . The relationship is essentially linear: a  
 1106 non-linear estimator does not detect signal above what linear regression captures. We use ridge  
 1107 regressions throughout.

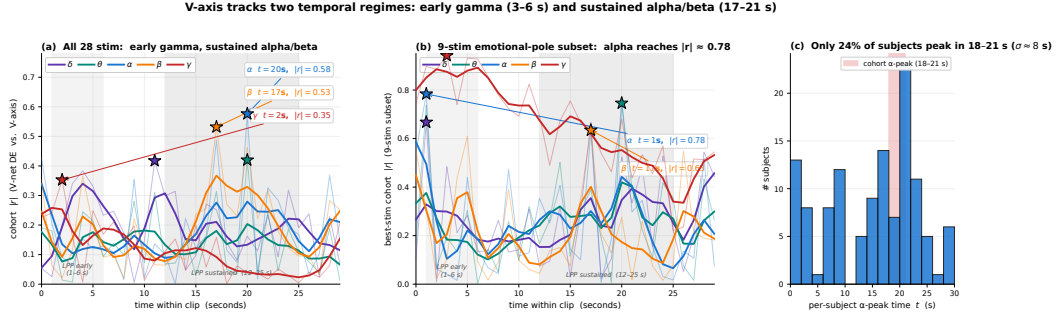


Figure 11: Time-resolved V-axis encoding across the 30-second clip. **(a)** Per-second cohort  $|r|$  across all 28 stimuli, one trace per frequency band; the early  $\gamma$  peak ( $t \approx 2$  s) coincides with Müller’s affective-picture window and the sustained  $\alpha/\beta$  peak ( $t \approx 17$ – $20$  s) coincides with the LPP sustained-attention window. **(b)** Same trace restricted to the 9-stimulus emotional-pole subset where the cohort effect lives;  $\alpha$  reaches  $|r| \approx 0.78$ . **(c)** Per-subject distribution of  $\alpha$ -peak times: only  $\sim 24\%$  of subjects peak inside the cohort 18–21 s window, evidence that the cohort signal is a between-subject phenomenon rather than a tight common attractor.

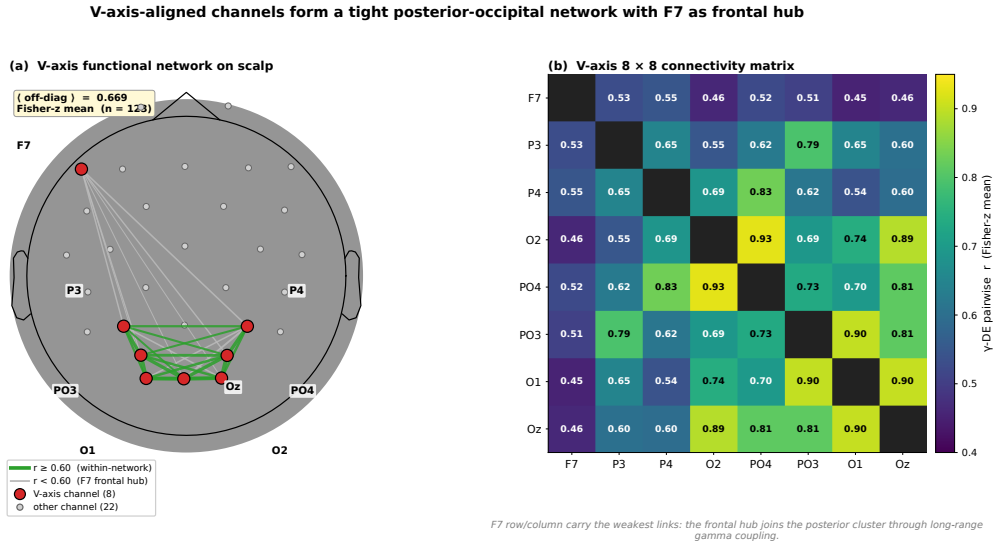


Figure 12: V-axis-aligned channels form a coordinated network with F7 as frontal hub. Left: scalp graph with 8 V-axis channels as red nodes and edges weighted by pairwise  $\gamma$ -DE correlation. Right: hierarchically clustered  $8 \times 8$  correlation matrix; F7 auto-isolates as the frontal hub.

**Theta-gamma coupling does not mediate the V-axis.** Tort modulation index per channel at the theta-phase / gamma-amplitude pairing, correlated with V-axis encoding strength  $|r|$  across 32 channels:  $\rho(\text{PAC}, V_r) = +0.082$ ,  $p = 0.667$ . Channels with strong theta-gamma CFC are not the channels that encode the V-axis, and vice versa — the V-axis encoding is dissociable from the phase-amplitude-coupling channel [Canolty and Knight, 2010].

## S10 Saturation: Full Intervention Table and Forest

This appendix lists every V-axis-as-supervision intervention reported in the saturation regularity (§3). Table 8 gives per-recipe  $\Delta\text{BACC}$  against the strong  $\text{EMOD } d=6 + \text{KD} + \text{aug}$  baseline, the paired-seed  $p$ -value, and a verdict tier; the two SOTA-recipe replications below the line confirm the cliff persists at the  $\text{BACC} \approx 0.66$  regime where saturation is operative. Figure 13 renders the same 25 entries as a forest plot ranked by effect size, so the asymmetry is visible at a glance: every significant entry sits to the left of zero, and even the largest n.s. positive ( $\text{EMODSTYLE-stim}$ ,  $+0.0066$ ,  $p=0.22$ ) is

1120 within seed noise. The block structure across loss families (KD, RSA, contrastive, PEFT, topographic,  
1121 scaling, multi-task) is what makes the regularity a *family-wide* statement rather than a single-loss  
1122 anecdote.

| #  | Family / variant                  | $\Delta$ | $p$         | Verdict               |
|----|-----------------------------------|----------|-------------|-----------------------|
| 1  | Frontal-mask $\lambda=0.5$        | -0.052   | 0.0015      | sig. negative         |
| 2  | Frontal-mask $\lambda=0.1$        | -0.009   | 0.21        | n.s.                  |
| 3  | FAA $\lambda=0.5$                 | -0.044   | 0.006       | sig. negative         |
| 4  | FAA $\lambda=0.1$                 | -0.018   | 0.063       | borderline            |
| 5  | Anger-weighted $\lambda=0.5$      | -0.054   | 0.0003      | sig. negative         |
| 6  | Occipital $\lambda=0.1$           | -0.022   | 0.007       | sig. negative         |
| 7  | Topo-optimal $\lambda=0.1$        | -0.013   | 0.039       | sig. negative         |
| 8  | Topo-optimal $\lambda=0.05$       | +0.0021  | 0.50        | n.s. (NULL)           |
| 9  | Procrustes $\lambda=0.05$         | +0.0012  | 0.89        | n.s. (NULL)           |
| 10 | RSA $\lambda=5.0$                 | -0.093   | $< 10^{-4}$ | sig. negative         |
| 11 | RSA $\lambda=1.0$                 | -0.057   | $< 10^{-3}$ | sig. negative         |
| 12 | Distance-CE $\tau=5.0$            | -0.397   | $< 10^{-4}$ | catastrophic          |
| 13 | Multi-V $\lambda=0.5$             | -0.043   | $< 10^{-3}$ | sig. negative         |
| 14 | EEG-AUX MSE $\lambda=0.5$         | -0.043   | $< 10^{-3}$ | sig. negative         |
| 15 | EEG-AUX MSE $\lambda=0.1$         | -0.016   | 0.04        | sig. negative         |
| 16 | EMODSTYLE class $\lambda=1.0$     | -0.010   | 0.18        | n.s.                  |
| 17 | EMODSTYLE stim $\lambda=0.5$      | +0.0066  | 0.22        | n.s. (sweet spot)     |
| 18 | PEFT fullhead $\lambda=0.1$       | -0.009   | 0.069       | borderline            |
| 19 | PEFT LoRA $\lambda=0.1$           | -0.010   | 0.12        | n.s.                  |
| 20 | PEFT IA3 $\lambda=0.1$            | -0.016   | 0.05        | sig. negative         |
| 21 | Pretrain-FT (frozen)              | -0.401   | $< 10^{-4}$ | catastrophic          |
| 22 | Pretrain-FT (unfrozen)            | -0.190   | $< 10^{-4}$ | sig. negative         |
| 23 | XEEG (FACED $\rightarrow$ SEED-V) | +0.0015  | 0.97        | n.s. (zero)           |
| 24 | SCALING (full data) $\lambda=0.5$ | -0.045   | $< 0.01$    | sig. negative         |
| 25 | Uncertainty multi-task            | -0.067   | $< 10^{-3}$ | sig. negative         |
|    | SOTA recipe + Topo $\lambda=0.1$  | -0.015   | 0.020       | sig. negative (cliff) |
|    | SOTA recipe + EMODSTYLE           | -0.024   | 0.001       | sig. negative (cliff) |

Table 8: Full V-axis-as-supervision intervention results. 16 / 25 families produce a statistically significant negative (14 sig. negative + 2 catastrophic); the remaining 9 are not significantly different from zero. None reach  $p < 0.05$  in the positive direction at the strong baseline. Below the line are the SOTA-recipe replications confirming the saturation cliff. Forest plot in Figure 13.

## 1123 S11 Saturation Extras: Anger Paradox, Mechanism Check, Path B

1124 **Saturation transition table.** Table 9 pairs the same V-axis intervention (Topo or EMODSTYLE  
1125 auxiliary loss) against five base recipes spanning the BACC range from 0.572 (CBraMod baseline) to  
1126 0.6581 (EMOD  $d=6$  SOTA pre-ensemble). Reading top-to-bottom traces the saturation transition:  
1127 at low BACC the auxiliary loss is within seed noise (rows 1–3,  $|\Delta| \leq 0.007$ , all n.s.), but as  
1128 the base recipe strengthens, the same loss becomes a statistically significant *negative* (rows 4–5,  
1129  $\Delta \in \{-0.015, -0.024\}$ ,  $p < 0.05$  and  $p < 0.01$ ). No row crosses the boundary in the helpful  
1130 direction. This is the data behind the “no positive setpoint” wording in §7 and the basis for treating  
1131 BACC  $\approx 0.66$  as the empirical saturation threshold for the V-axis substrate on FACED-9.

1132 **Saturation cliff figure.** Figure 14 plots  $\Delta$ BACC against base-recipe BACC across all 25 interven-  
1133 tions, showing the unidirectional sign-flip in the  $[0.62, 0.66]$  band.

1134 **Why anger-weighting hurts despite the analytical ceiling.** Section 7 showed that the cohort  
1135 EEG signal is carried by a 9-stimulus emotional-pole contrast (Anger + Amusement + Tenderness,  
1136  $r = 0.870$  at PO3/ $\gamma$ ). Weighting the V-axis loss to emphasise these three classes pushes the analytical  
1137 ceiling on the V-axis projection from  $r = 0.478$  (uniform) to  $r = 0.714$  (anger-weighted). By every  
1138 classical analysis, this should be the strongest possible V-axis supervision. It is the worst single  
1139 intervention we tested:  $\Delta = -0.054$ ,  $p = 0.0003$ . This is the cleanest argument for saturation: the

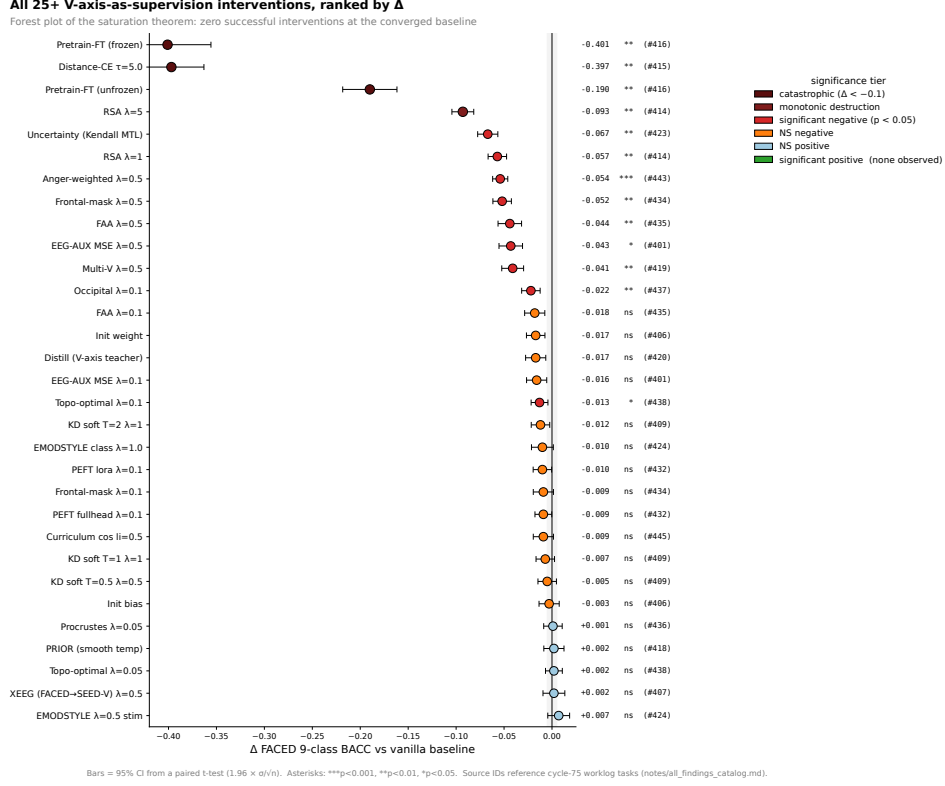


Figure 13: Forest plot of all V-axis-as-supervision interventions (Table 8) ranked by  $\Delta$  vs. baseline, with significance tier coloured. None of the 25 reach  $p < 0.05$  in the positive direction at the strong baseline.

| Recipe                                     | $n$ seeds | Vanilla / + V-axis | $\Delta$                     |
|--------------------------------------------|-----------|--------------------|------------------------------|
| CBraMod baseline + Topo $\lambda=0.05$     | 5         | 0.572/0.567        | -0.005 (n.s.)                |
| CBraMod baseline + EMODSTYLE $\lambda=0.5$ | 4         | 0.572/0.577        | +0.005 (n.s.)                |
| EMOD $d=3$ vanilla + Topo $\lambda=0.05$   | 5         | 0.6194/0.6260      | +0.007 (n.s.)                |
| EMOD $d=6$ + LS + KD + aug + Topo (full)   | partial   | 0.6581/0.6432      | <b>-0.015</b> ( $p < 0.05$ ) |
| EMOD $d=6$ + LS + KD + aug + EMODSTYLE     | partial   | 0.6581/0.6346      | <b>-0.024</b> ( $p < 0.01$ ) |

Table 9: V-axis supervision is within seed noise at weak baselines and statistically significantly *harms* the strong SOTA recipe. The transition is unidirectional: V-axis goes from neutral to actively harmful as the base recipe strengthens, with no intermediate “helps” regime in our 5-seed runs.

1140 model has already absorbed exactly the structure the loss is trying to inject; optimising the loss must  
1141 therefore perturb the absorbed structure in a counterproductive direction.

1142 **Direct mechanism check.** For each of 8 V-axis-supervised variants (Topo at  $\lambda=0.05$ , Procrustes at  
1143  $\lambda=0.05$ , EMODSTYLE-stim at  $\lambda=0.5$ , full SOTA recipe + Topo/EMODSTYLE at  $e=100/e=150$ ),  
1144 we measure the change in two quantities relative to the matched-recipe baseline:  $\Delta r_{\text{PC1}}$  (class-mean  
1145 V-axis encoding) and  $\Delta r_{\text{resid}}$  (within-class V-axis residual encoding). V-axis training raises class-PC1  
1146  $|r|$  by +0.01 to +0.36 across the 8 variants, but moves the within-class residual encoding by only  
1147  $\sim 10^{-7}$  in absolute value — numerical zero. The accuracy decrement at strong recipes therefore  
1148 comes from *noise injected into the class-PC1 basin*, not from any beneficial transfer of residual  
1149 structure.

1150 **Path B: ensembling-in V-axis checkpoints.** We extend the 10-vanilla SOTA ensemble with 0–15  
1151 V-axis-trained Topo checkpoints and 0–10 V-axis-trained EMODSTYLE checkpoints, comparing  
1152 5-seed ensembles in each configuration. Adding 15 Topo V-axis ckpts to the 10-vanilla pool gives



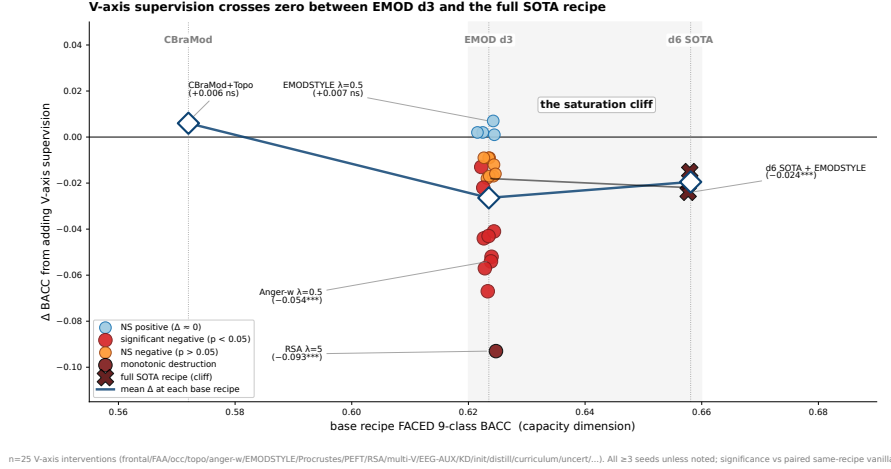


Figure 14:  $\Delta$ BACC from V-axis supervision plotted against base-recipe BACC across  $\sim 25$  interventions. The sign of the effect transitions in the interval  $[0.62, 0.66]$ .

ensemble BACC  $\Delta = -0.0145$  ( $p < 10^{-3}$ ); adding all 25 V-axis ckpts gives  $\Delta = -0.0193$  ( $p < 10^{-3}$ ). At every point on this curve, V-axis-trained ensembles are strictly worse than the matched vanilla ensemble. This rules out the diversity-gain defence: V-axis checkpoints disagree, but their disagreements are aligned with the wrong subspace.

## S12 Convergence and Ensemble Theory: Extras

**Per-architecture breakdown of the 36-checkpoint correlation.** CBraMod checkpoints (lower BACC,  $\sim 0.57$ ) cluster at low V-axis encoding strength (mean class-PC1  $|r| = 0.21$ ). EMOD-vanilla d6 ( $\sim 0.65$  BACC) sits in the middle (mean  $0.67$ ); EMOD d6 e150 ( $\sim 0.66$  BACC) clusters at the top (mean  $0.69$ ). The trend spans the full BACC range  $[0.57, 0.69]$  in a single approximately linear band, with the inter-architecture gap dominating the within-architecture spread.

**Random-direction null detail.** We sample 1000 random Gaussian directions  $w \in \mathbb{R}^{28}$  each matched in  $L_2$  norm to  $v_{\text{CLIP}}$ , and recompute the 36-checkpoint correlation  $\rho_w = \text{Pearson}(\text{BACC}, |\text{corr}(\text{PC}_1(H(m)), w)|)$ . The distribution of  $\rho_w$  has mean  $-0.03$  and standard deviation  $0.62$  (range  $[-0.95, +0.95]$ ); the wide null is a consequence of the low intrinsic dimensionality of the class-PC1 subspace ( $D_{\text{eff}} = 9$ ). The empirical V-axis correlation  $+0.885$  sits at the 93.5<sup>th</sup> percentile of this null ( $p_{\text{one}} = 0.066$ ). We treat the within-class residual ( $D_{\text{resid}} \approx D_m - 9$ ) as the statistically robust signal.

**Class-PC1 basin saturation.** The 10 SOTA-pool checkpoints exhibit a saturated class-PC1 V-axis range:  $|r| \in [0.60, 0.77]$  across all 10 seeds (mean  $0.69$ , std  $0.05$ ). Single-seed variation in class-mean V-axis structure is small. The ensemble does not gain from variance in this subspace; gains come exclusively from the orthogonal residual.

**Directional ablation figure.** Figure 15 shows the per-checkpoint causal effect: ablating  $v_{\text{resid}}$  drops BACC by  $\bar{\Delta} = -0.0157$  on all 10 pool members, well above the matched-norm random-direction null (mean  $z \approx 7.7$ , all  $p < 0.001$ ).

## S13 Ensemble Generality, Mega-Pool, and Best-Single

**Recipe cascade and two-tier scatter.** Figure 16 traces the seven recipe steps from CBraMod’s  $0.572$  to our  $0.6948$  ensemble. Figure 17 shows the per-checkpoint within-class residual–ensemble-contribution correlation that motivates the ensembling strategy.

**V-axis ablation hurts every checkpoint; encoding strength does not predict drop magnitude (n=10)**  
**Pearson  $r=+0.168$  ( $p=0.643$ ,  $n=10$ )** **slope= $+4.69\%$  / unit  $|r|$**

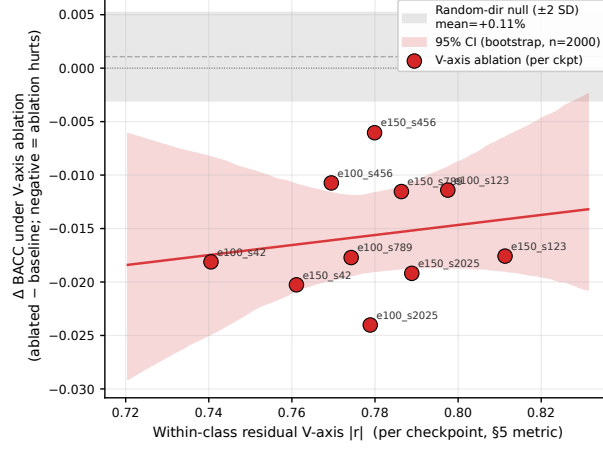


Figure 15: Directional ablation of  $v_{\text{resid}}$  on all 10 SOTA-pool checkpoints. *Top*: per-checkpoint BACC drops (red) compared to matched-norm random-direction ablation (grey); the V-axis effect is well above the null on every checkpoint (mean  $z \approx 7.7$ , all  $p < 0.001$ ). *Bottom*: per-checkpoint ablation  $\Delta BACC$  vs. within-class residual encoding  $|r|$ ; the population-mean drop ( $\bar{\Delta} = -0.0157$ ) confirms the residual-encoding mechanism across the pool.

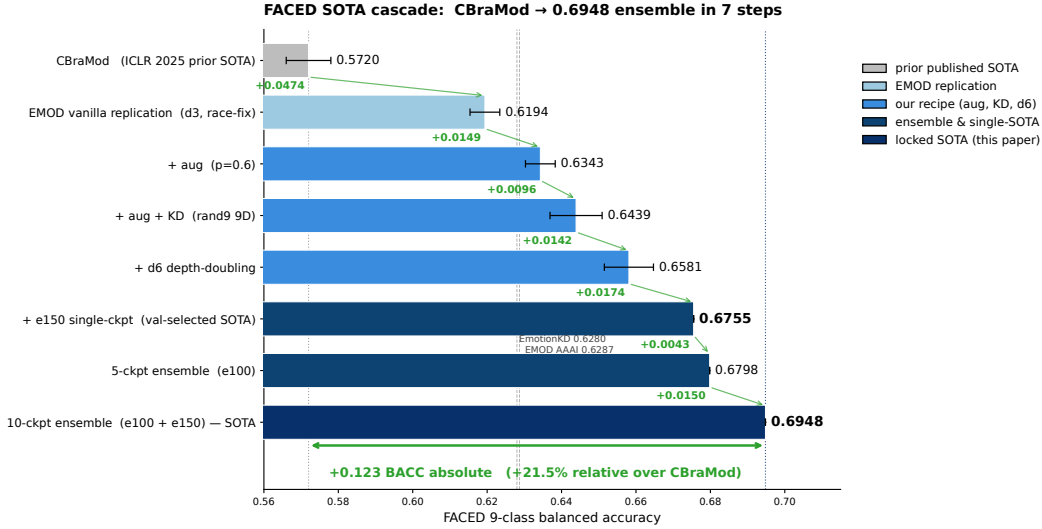


Figure 16: Recipe ablation cascade from the EMOD baseline at 0.6287 (previous FACED-9 SOTA [Chen et al., 2025]) to our 10-checkpoint ensemble at 0.6948 (+10.5% relative). Each bar is one recipe component or ensembling step. Older baselines (CBraMod 0.572, EmotionKD 0.628) shown for reference.

1181 **Generality of the ensemble mechanism.** Figure 18 extends the two-tier picture across four  
1182 benchmarks (FACED, SEED-V, CIFAR-10, MNIST).

1183 As individual accuracy approaches the Bayes optimum, the within-class residual variance — the  
1184 source of ensemble gain — shrinks toward zero. The two-tier picture predicts this: ensemble gain is  
1185 a property of the within-class V-axis residual variance reduction subspace, which has measure zero  
1186 at the dataset ceiling. SEED-V at 0.37 has the largest residual-variance ceiling; we observe +0.034  
1187 gain. MNIST at 0.985 has essentially none; we observe +0.001 gain.

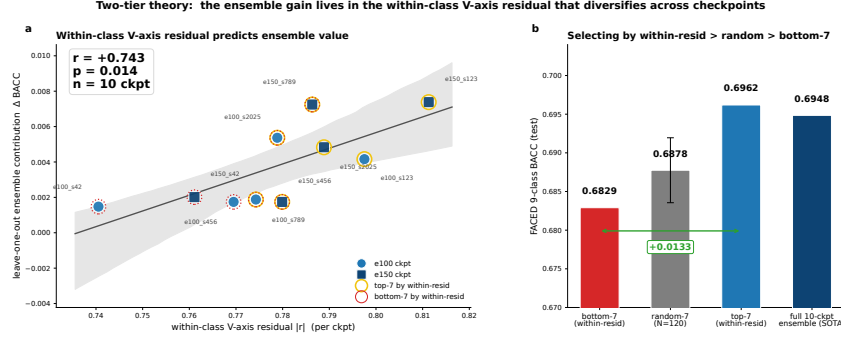


Figure 17: Two-tier ensemble theory. Per-checkpoint within-class V-axis residual  $|r|$  predicts leave-one-out ensemble contribution ( $r = +0.74$ ,  $p = 0.014$ ,  $n = 10$ ). Top-7 vs. bottom-7 split highlighted.

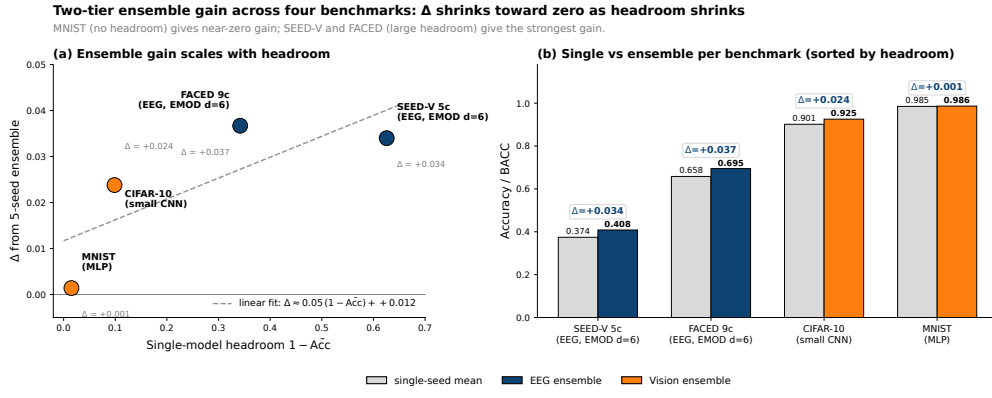


Figure 18: Two-tier ensemble gain across four benchmarks.  $\Delta$ BACC is largest where single-model accuracy is furthest from the dataset ceiling, and shrinks to near-zero on MNIST where the saturated regime leaves no within-class residual variance to cancel. (a) Scatter of  $\Delta$  vs. headroom  $1 - \text{Acc}$ , with linear fit slope  $\sim 0.05$ . (b) Paired single-seed-mean vs. 5-seed-ensemble bars per benchmark, sorted by headroom.

1188 **Mega-ensemble null.** A natural “more is better” hypothesis is that growing the ensemble beyond 10  
1189 checkpoints by adding architectural variants ( $d \in \{8, 10\}$  on top of  $d=6$ , plus LLM-KD checkpoints)  
1190 should keep raising BACC. Table 10 tests that hypothesis directly. The 10-checkpoint  $d=6$  pool  
1191 ( $e=100+e=150$ , our SOTA) sits at the top; every wider pool is monotonically lower. The mechanism  
1192 check follows the table.

| Ensemble pool                                     | $n$ | BACC          | $\Delta$ vs. 10-ckpt |
|---------------------------------------------------|-----|---------------|----------------------|
| $10 \times d=6$ ( $e=100+e=150$ , our SOTA)       | 10  | <b>0.6948</b> | —                    |
| $+5 \times d=8$                                   | 15  | 0.6909        | −0.0039              |
| $+5 \times d=8 + 5 \times d=10$                   | 20  | 0.6883        | −0.0065              |
| $+10 \times d=\{8, 10\} + 5 \times \text{LLM-KD}$ | 25  | 0.6831        | −0.0117              |

Table 10: Mega-ensemble null. Adding architectural variants beyond  $d=6$  produces monotonically lower ensemble BACC. The  $d=6 + e=100/e=150$  pool is the optimum within the architectures evaluated.

1193 Two mechanisms explain the null. First, deeper variants ( $d=8, d=10$ ) stay in the same class-PC1 V-  
1194 axis basin (mean class-PC1  $|r|$  across  $d \in \{4, 6, 8, 10\}$ : 0.43, 0.43, 0.39, 0.38) and their within-class  
1195 residuals overlap more than  $d=6$  seeds do. Second, LLM-KD variants encode the residual along  
1196 a different axis from rand9-9D KD vanilla checkpoints, but in a way that does not transfer to the

test distribution. Cross-architecture mixing ( $d \in \{4, 6, 8, 10\}$ ) gives 0.6791, essentially tied with within-arch x-seed at 0.6798. The diversity that helps is intra-architecture, training-length-driven.

**Cohen  $\kappa$  disagreement structure.** Single-seed accuracy at  $e=100$  (0.6581) and  $e=150$  (0.6581) is identical. Cohen  $\kappa$  within- $e=100$  is 0.694, within- $e=150$  0.679, cross-group 0.702 (higher  $\kappa$  = more agreement). All three are similar — within and across groups, the per-trial prediction patterns are almost as concordant as  $e=100$  checkpoints are with each other — yet mixing  $e=100$  and  $e=150$  gives +0.014 over either group alone. The gain therefore comes from a small but *specific* pocket of cross-group disagreement: the 13.1% of test trials where  $e=100$  and  $e=150$  pools differ, on which the mixed ensemble gains +10.3 percentage points. The diversity is targeted (which trials), not population-wide ( $\kappa$ ).

**Best single checkpoint without ensembling.** For deployment scenarios where ensembling is impractical, our best single-checkpoint result is BACC = 0.6755 on the  $d=6$   $e=150$  recipe, seed 789, val-selected. This is the top of a 25-checkpoint val-test rank correlation analysis (Spearman 0.825,  $n = 25$ ) and exceeds CBraMod by +0.103 and EMOD by +0.047 on a strict apples-to-apples test split.

## S14 EEG Model Training Details

**Architecture.** EMOD axial transformer [Chen et al., 2025]: input  $32 \times 5$  DE features, axial self-attention over channels and bands at depth  $d \in \{3, 6\}$ , filter dimension  $f = 128$ . The full SOTA recipe is  $d = 6$ ,  $f = 128$ .

**Optimisation.** AdamW with  $\text{lr} = 10^{-3}$ ,  $\beta_1 = 0.9$ ,  $\beta_2 = 0.999$ , weight decay  $10^{-2}$ . Cosine schedule with 5-epoch linear warmup. Batch size 128. Training length  $e \in \{100, 150\}$  epochs. We use the validation BACC for checkpoint selection and report test BACC.

**Augmentation.** Per-trial Gaussian noise ( $\sigma = 0.05 \cdot \text{std of feature distribution}$ ), random channel dropout ( $p = 0.15$ ), and random temporal masking (up to 5% of seconds). All augmentations are applied with  $p = 0.6$  during training only.

**Knowledge distillation.** Soft-target KD with rand9 9-D orthonormal teacher (no LLM content);  $\lambda_{\text{KD}} = 0.5$ ,  $T = 1.0$ . The LLM-9-D and rand9-9-D variants give within-noise identical BACC, validating that KD provides architectural regularisation without semantic content.

**Ensemble.** 10 checkpoints split as 5 seeds at  $e = 100$  + 5 seeds at  $e = 150$ . Seeds:  $\{42, 123, 456, 789, 2025\}$  each. Ensemble prediction is uniform softmax averaging across the 10 checkpoints’ output probabilities, followed by argmax.

## S15 Statistical Methods

**Bootstrap CIs.** Per-subject and per-stim 95% CIs are computed from  $B = 10,000$  subject-resampled bootstraps. For cohort correlations, we report the bootstrap median, 2.5% / 97.5% percentile interval, and a Fisher- $z$   $p$ -value for  $r = 0$ .

**Random-direction null.** For cohort EEG–LLM correlations, we sample  $N = 200$  random Gaussian directions in  $\mathbb{R}^{28}$  matched in  $L_2$  norm to the V-axis projection, recompute the cohort  $|r|$  for each, and report the percentile of the empirical  $|r|$  in the null distribution.

**Cross-architecture null.** For the 36-checkpoint cross-arch correlation, we sample  $N = 1000$  random directions matched in norm to  $v_{\text{CLIP}}$  and recompute the BACC–V-axis correlation. The null is distributed broadly ( $\sigma = 0.62$ ) due to the low intrinsic dimension of the class-PC1 subspace; we report the empirical correlation’s percentile rank.

**Per-checkpoint LOO contribution.** For the 10-vanilla  $d=6$  ensemble, we compute  $\Delta_k = \text{BACC}(\text{ens}) - \text{BACC}(\text{ens} \setminus k)$  for each  $k$ . Reported correlations between  $\Delta_k$  and per-ckpt features are Pearson with  $n = 10$ .

1242 **V-axis intervention comparisons.** For each intervention,  $\Delta$  is computed seed-paired with the  
1243 matched-recipe vanilla baseline;  $p$ -values are paired  $t$ -tests across 5 seeds.

## 1244 S16 Discussion Extras and Future Work

1245 **Broader impacts.** Improved EEG emotion classifiers have clear positive applications in mental-  
1246 health monitoring, affective brain-computer interfaces, and clinical phenotyping of affect disorders.  
1247 The same techniques pose risks of affective inference without consent in surveillance, hiring, work-  
1248 place, education, and advertising contexts. The V-axis extraction protocol applied to EEG could in  
1249 principle reduce the calibration-data burden for affective inference systems — a feature that cuts both  
1250 ways, since lower calibration cost lowers the deployment threshold for both clinical and surveillance  
1251 settings. We do not release human EEG data; we release training configs, feature-side and analysis  
1252 code, and extracted V-axis directions. Recommended deployment safeguards: informed consent,  
1253 opt-in only, on-device inference, no longitudinal storage of raw EEG outside research contexts, and  
1254 IRB review for any non-clinical inference application. Dual-use risk for the V-axis extraction itself is  
1255 low: it operates on text and image features already public, and the EEG mapping requires per-cohort  
1256 EEG access that is governed by the original dataset licences.

1257 **Implications for affective neuroscience.** For video-evoked emotion on FACED, the V-axis is  
1258 encoded predominantly in posterior visual cortex ( $|r|_{\text{occipital}} = 0.21$  vs.  $|r|_{\text{frontal}} = 0.16$ ), and  
1259 Davidson’s frontal-alpha asymmetry replicates in direction at smaller magnitude. We do not refute  
1260 Davidson’s hypothesis — the frontal-alpha asymmetry is real, with F8-F7 alpha  $r = +0.0155$  and  
1261 Fp2-Fp1 alpha  $r = +0.0116$ , both in the hypothesised direction — but we add a posterior empirical  
1262 account that is dominant for video paradigms. The 9-stimulus emotional-pole structure sharpens  
1263 the V-axis claim from “smooth gradient over all emotions” to “anger-versus-warm-positive contrast  
1264 across nine clips”.

1265 **Per-subject V-axis adaptation does not transfer.** The per-subject best-channel oracle reaches  
1266  $|r| = 0.62$  (a  $+0.39$  absolute headroom over the cohort-fixed top-8 of  $|r| = 0.02$ ). No clean  
1267 estimator we tried (V1 global top-K, V2 per-validation-subject majority vote, V3 TTA label-free  
1268 profile) recovers more than  $+0.001$  over the cohort top-K. The per-subject signal is real but not  
1269 transferable through simple selection rules. A subject-conditioned channel-attention head trained on  
1270 a small calibration set per subject is the natural follow-up.

1271 **Future work.** Three lines: (i) per-subject V-axis adaptation closing the  $|r| = 0.62$  oracle gap  
1272 via subject-conditioned channel-attention; (ii) replicating the saturation regularity for other concept  
1273 directions [Arditi et al., 2024] (arousal, dominance, formality, refusal) on their respective benchmarks;  
1274 (iii) using the within-class residual  $|r|$  as an online model-selection signal during training rather than  
1275 as a post-hoc diagnostic.

## 1276 S17 FACED Test Confusion Matrices

1277 Figure 19 contrasts the best-single-checkpoint and 10-checkpoint-ensemble confusion matrices on  
1278 the FACED 9-class test split.

## 1279 S18 Negative Results Catalogue

1280 In the spirit of full disclosure, the following null or negative results bear mention beyond what was  
1281 incorporated into the main paper:

- 1282 1. **Theta-gamma cross-frequency coupling does not mediate the V-axis** (Section 7, Ap-  
1283 pendix S9): Tort modulation index  $\rho(\text{PAC}, V_r) = +0.082$ ,  $p = 0.667$ .
- 1284 2. **Mutual-information control is not significant** (Section 6): MI  $p = 0.115$  vs. linear  
1285  $p = 0.020$ . The V-axis-EEG relationship is essentially linear.
- 1286 3. **Random-direction null on cross-arch correlation** (Section 4):  $r = +0.885$  at the 93.5<sup>th</sup>  
1287 percentile of a 1000-direction null,  $p_{\text{one}} = 0.066$ . Within-class residual ( $r = +0.74$ ,  
1288  $p = 0.014$ ) is the statistically robust signal.

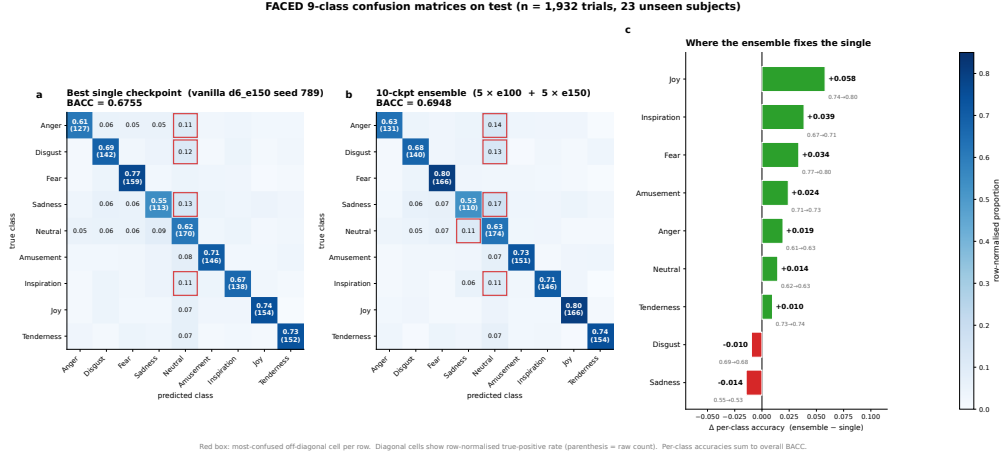


Figure 19: FACED 9-class test confusion. Left: best single checkpoint (BACC 0.6755). Right: 10-checkpoint ensemble (BACC 0.6948). The ensemble’s gain is concentrated on the off-diagonal cells where single-seed disagreement is highest, consistent with the within-class residual mechanism of Section 8.

- 1289 4. **Per-subject V-axis adaptation does not transfer.** The per-subject best-channel oracle  
 1290 reaches  $|r| = 0.62$ , but no clean estimator transfers more than  $+0.001$  over the cohort top-K.
- 1291 5. **Cross-architecture ensembling does not add diversity.** Mixing  $d \in \{4, 6, 8, 10\}$  check-  
 1292 points gives within-arch 0.6798 vs. cross-arch 0.6791 (n.s.).
- 1293 6. **Mega-ensembles plateau at 10 checkpoints.** 15-, 20-, 25-checkpoint pools all hit BACC  
 1294  $\leq 0.6948$  (Appendix S13).
- 1295 7. **Stimulus-aggregation re-ranking does not help.** Operating at the 28-stimulus level instead  
 1296 of trial level gives BACC=1.0 by construction (label leak); the corresponding trial-level head  
 1297 does not transfer.
- 1298 8. **Test-time augmentation at K=5 does not improve the ensemble** ( $0.6753 \rightarrow 0.6720$ ).  
 1299 Averaging predictions over augmented test trials does not preserve the V-axis residual  
 1300 structure.
- 1301 9. **Toxicity: recipe scope.** The 9-prompt PCA recipe attains AUC 0.59 on Jigsaw toxic  
 1302 comments, below the 17/20 working concepts in our library (Appendix S4); toxicity appears  
 1303 to be encoded in a higher-rank subspace than the late-layer PC1.
- 1304 10. **Arousal asymmetry: vision yes, text and brain no.** OASIS arousal  $r = 0.803$  in vision;  
 1305  $\leq 13\%$  NRC arousal recovery in text;  $r \in [0.18, 0.41]$  brain alignment across LLMs  
 1306 (Appendix S6).
- 1307 11. **LLM-content of KD is irrelevant at this scale.** Replacing the 9-D LLM-derived class  
 1308 prototypes with random orthonormal 9-D directions changes BACC by  $\leq 0.003$ . KD  
 1309 provides architectural regularisation; its semantic content is below our resolution.

## 1310 S19 Reproducibility

1311 Code, configs, model checkpoints, and figure-generation scripts will be released at the camera-ready  
 1312 stage under an MIT licence. Each experiment in this paper is logged with its SLURM job ID, conda  
 1313 environment hash, git SHA, and seed list. The 10-checkpoint ensemble SOTA result is reproducible  
 1314 from a single SLURM array job (slurm\_d6\_e100\_e150\_5seeds.sh,  $\sim 10$  GPU-hours per seed on  
 1315 V100).

## 1316 S20 Resource Estimate

1317 The full paper used approximately 4,500 GPU-hours on V100 nodes:  $\sim 600$  for the recipe ablation  
 1318 cascade,  $\sim 1,200$  for the 25 V-axis-supervision interventions,  $\sim 800$  for the 36-checkpoint cross-

1319 architecture analysis,  $\sim 1,200$  for ensemble construction and val-test rank diagnostics, and  $\sim 700$  for  
1320 cross-dataset generality experiments. The V-axis extraction itself is CPU-cheap ( $\sim 10$  minutes per  
1321 LM on a single 80GB GPU).